



# A reliable and explainable EEG-based eye state classification via XGBoost with missing data resilience and SHAP-LIME explanations

Hamzah Dabool <sup>a</sup>, Zaher Aghbari <sup>b</sup>, Naveed Ahmed <sup>b</sup>, Imad Afyouni <sup>b</sup>, Hany Alashwal <sup>c</sup>, Shifan Khanday <sup>d</sup>

<sup>a</sup> Dubai Medical University, Dubai, United Arab Emirates

<sup>b</sup> Department of Computer Science, University of Sharjah, Sharjah, United Arab Emirates

<sup>c</sup> Big Data Analytics Center, College of Information Technology, United Arab Emirates University, Al Ain, Abu Dhabi, United Arab Emirates

<sup>d</sup> College of Medicine, Dubai Medical University, Dubai, United Arab Emirates

## ARTICLE INFO

### Keywords:

Eye state classification  
EEG  
Brain–computer interface  
XGBoost  
Significance testing  
Misclassification error analysis  
Feature importance  
SHAP  
LIME  
Missing data resilience

## ABSTRACT

This study presents an interpretable, electrode-domain classification framework for Electroencephalography (EEG) eye-state detection using eXtreme Gradient Boosting (XGBoost), evaluated on the UCI EEG Eye State dataset. The approach operates directly on sensor-level voltage time series, bypassing handcrafted spectral or temporal feature extraction, making it suitable for low-latency and physiologically interpretable analysis. Eye-state discrimination from scalp EEG is challenging due to inter-subject variability, low signal-to-noise ratio (SNR), and spatially overlapping neural correlates of eye-open and eye-closed states. The proposed framework was benchmarked against conventional classifiers using rigorous validation protocols (3-, 5-, 7-, and 10-fold cross-validation and an 80/20 hold-out split), achieving 95.73% accuracy and 99.13% Area Under the Curve (AUC), with statistical significance confirmed via t-test, Wilcoxon, and Friedman tests ( $p < 0.001$ ). A comprehensive error–signal analysis, encompassing decision ambiguity, calibration reliability, and sample-wise hardness, showed that most misclassifications occurred in regions of high intra-class variance and inter-class similarity. Calibration curves demonstrated well-aligned predicted probabilities, supporting confidence-aware deployment. Explainability analysis using SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), and Gini importance consistently identified frontal (F7), parietal (P7), and occipital (O1) electrodes as key contributors, aligning with known neurophysiological mechanisms related to visuomotor processing and alpha-rhythm modulation during eye closure. Robustness was evaluated under random and targeted electrode dropout (1%–20%). Performance remained above 87.4% accuracy (AUC > 94.6%) with 20% missing channels, demonstrating graceful degradation.

## 1. Introduction

Electroencephalography (EEG) is a safe, non-invasive method for recording brain activity through electrodes placed on the scalp, providing real-time insights into neural dynamics [1,2]. It is widely applied in diagnosing epilepsy, sleep disorders, and brain injuries, as well as in studies of attention, memory, and emotions [3]. EEG signals are acquired from standardized scalp locations [4]. In practical neuroscience-oriented applications, low- to moderate-density EEG configurations are commonly preferred due to their reduced setup complexity and lower susceptibility to volume conduction effects.

Eye state classification aims to determine whether a person's eyes are open or closed using physiological signals, most commonly EEG [5].

Because eye state is closely linked to alertness, attention, and drowsiness, it plays an important role in safety-critical and clinical applications. For example, in driver monitoring systems, detecting prolonged eye closure can help prevent accidents caused by fatigue, while in brain–computer interface (BCI) systems, eye state awareness can improve human–machine interaction [6].

EEG-based eye state detection is particularly valuable in medical contexts, where it provides a reliable means of monitoring alertness, sleep-related behavior, and overall brain activity [7]. It is commonly employed in sleep studies to distinguish between eyes-open and eyes-closed conditions, supporting the diagnosis of disorders such as insomnia and sleep apnea [8]. Importantly, resting EEG with eyes closed differs fundamentally from sleep EEG: while sleep EEG is dominated

\* Corresponding author.

E-mail addresses: [hamzah.dabool@dmu.ae](mailto:hamzah.dabool@dmu.ae) (H. Dabool), [zaher@sharjah.ac.ae](mailto:zaher@sharjah.ac.ae) (Z. Aghbari), [nahmed@sharjah.ac.ae](mailto:nahmed@sharjah.ac.ae) (N. Ahmed), [iafyouni@sharjah.ac.ae](mailto:iafyouni@sharjah.ac.ae) (I. Afyouni), [halashwal@uaeu.ac.ae](mailto:halashwal@uaeu.ac.ae) (H. Alashwal), [dr.shifan@dmu.ae](mailto:dr.shifan@dmu.ae) (S. Khanday).

<https://doi.org/10.1016/j.bspc.2026.110161>

Received 26 July 2025; Received in revised form 25 February 2026; Accepted 18 March 2026

Available online 30 March 2026

1746-8094/© 2026 Published by Elsevier Ltd.

by stage-specific oscillatory patterns, resting closed-eye EEG is characterized by increased alpha activity without equivalent state-dependent dynamics [9]. Unlike camera-based methods, EEG does not depend on visual conditions, making it robust to lighting changes and subject movement in clinical environments.

EEG-based eye state classification has been studied using a broad range of classifiers, including linear models, instance-based methods, ensemble learners, and deep learning architectures, each offering different trade-offs between accuracy, interpretability, and data requirements [10]. In this work, eXtreme Gradient Boosting (XGBoost) was selected because it captures nonlinear inter-electrode relationships, operates directly on electrode-domain signals, and natively handles missing values—a common challenge in EEG acquisition. Compared to deep learning approaches, it requires fewer samples and avoids complex feature engineering. To ensure interpretability, SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) were employed as complementary explanation methods, enabling both global electrode-level attribution and local instance-specific insight into model decisions.

Well-established surface EEG studies frequently rely on frequency-domain, coherence-based, or complexity-driven representations combined with conventional ensemble classifiers, as demonstrated in prior work on emotion regulation and neural sound encoding [11]; however, such approaches require access to continuous EEG signals, which are not available in the pre-structured electrode-domain dataset used in this study.

However, existing approaches to EEG-based eye state detection face notable limitations. Most prior studies focus primarily on improving accuracy, while paying little attention to robustness under missing or corrupted EEG data and offering only limited interpretability of model decisions. These gaps restrict the reliability and clinical trust of current models, creating the need for methods that combine fault tolerance with transparent explanations. To the best of our knowledge, few studies have systematically evaluated eye state classification on the UCI EEG Eye State dataset using eXtreme Gradient Boosting (XGBoost) while also addressing missing-data resilience and incorporating both global (dataset-wide) and local (instance-specific) explainability with SHAP and LIME.

To address these gaps, this study sets out four main objectives: (1) to benchmark XGBoost against traditional baseline classifiers; (2) to evaluate its robustness under different missing-data scenarios, including random and targeted deletion, without requiring imputation; (3) to provide a comprehensive explainability analysis of electrode contributions through SHAP and LIME; and (4) to outline the limitations of the present work and identify future research directions.

The outcomes of this study have direct implications for application domains where reliability and interpretability are critical. In healthcare settings, the model's robustness to missing or corrupted EEG channels supports its use in clinical monitoring environments. For research and diagnostic applications, the integration of SHAP and LIME provides transparent, electrode-level explanations that enhance trust and facilitate expert interpretation. In safety-critical systems such as driver monitoring, maintaining stable performance without the need for data imputation enables dependable detection of drowsiness and reduced alertness. Collectively, these results demonstrate the broader applicability of EEG-based eye state detection beyond controlled experimental conditions.

## 2. Related work

Early work applied classical machine learning models to the UCI EEG Eye State dataset. Koklu and Sabancı [12] evaluated  $k$ -nearest neighbor (kNN) and multilayer perceptron (MLP) classifiers using WEKA (Waikato Environment for Knowledge Analysis) implementations without preprocessing or feature transformation. The kNN model, based on Euclidean distance and neighborhood sizes  $k = 1$ –10, achieved

its best performance at  $k = 3$  with an accuracy of 84.06%, Mean Absolute Error (MAE) of 0.1993, and Root Mean Square Error (RMSE) of 0.3465. In contrast, the MLP — configured with a single hidden layer — reached a maximum accuracy of 56.46% with seven neurons. Although this comparison demonstrates sensitivity to model choice and basic hyperparameter tuning, the study did not specify its validation strategy, omitted key learning parameters (e.g., optimizer, learning rate, activation functions, epochs), reported no computational metrics, and assumed fully observed EEG data without addressing missing channels, sensor noise, or real-world robustness.

A comprehensive benchmarking study by Rösler and Suendermann [10] evaluated 42 machine learning classifiers within the WEKA framework, including instance-based (KStar, IB1), rule- and tree-based (JRip, J48), probabilistic (Naïve Bayes), linear (logistic regression), support vector machines, and neural network models. All classifiers were assessed using stratified ten-fold cross-validation on 14,977 samples (after removal of three erroneous instances), with class proportions of 55.12% eye-open and 44.88% eye-closed. The instance-based KStar classifier achieved the lowest classification error of 2.7% (approximately 97.3% accuracy), substantially outperforming the majority-vote baseline error of 44.9%. Performance was further improved by tuning the global blend parameter from  $b = 20$  to approximately  $b = 40$ , although hyperparameters for other models were not reported. Despite its strong accuracy, KStar required over 38 min to process the dataset on a virtualized dual-core system with 32 GB RAM, highlighting a significant trade-off between predictive performance and computational practicality. Robustness analysis was limited to extreme outlier testing, with no systematic evaluation of missing data, noise, deployment constraints, or explainability.

Hybrid and ensemble-based approaches have been proposed to balance predictive accuracy and computational efficiency. Nilashi et al. [13] introduced a two-stage framework combining Learning Vector Quantization (LVQ) with bagged tree ensembles and evaluated it against several classical classifiers, including bagged trees, CART, LDA, Random Trees, Naïve Bayes, and MLP. The data were first clustered into eight LVQ-based groups (four per class) using a learning rate of 0.005, Euclidean distance, and variance-based normalization, followed by bagged tree classification within each cluster. Performance was assessed using stratified five-fold cross-validation, with results reported in terms of accuracy and ROC-AUC. The hybrid LVQ+bagged tree model achieved a mean accuracy of 94.31% (maximum 95.34%) and an AUC of 0.9628, outperforming all competing methods. The study also reported real-time feasibility, achieving a throughput of 58,942 predictions per second on an Intel Core i7-6700HQ system with 8 GB RAM. However, despite claims of inherent interpretability, no explicit feature attribution was provided, robustness to missing channels, sensor noise, or cross-subject variability was not evaluated, and statistical significance was assessed only via the Friedman test with Holm post hoc correction ( $p < 0.05$ ).

Deep learning approaches have emphasized real-time EEG-based eye state recognition using optimized neural architectures and diverse validation strategies. Reddy and Behera [14] evaluated supervised MLP/Feed-Forward Neural Networks (FFNN), unsupervised deep belief networks (DBNs), and dropout-regularized deep neural networks for online BCI applications. Performance was assessed using multiple validation schemes, including stratified holdout splits (80%/10%/10% and 80%/20%), ten-fold cross-validation, and leave-one-out cross-validation (LOOCV). MLP models with a  $14 \times 40 \times 2$  architecture achieved accuracies between 93.0% and 94.92%, while LOOCV yielded a peak accuracy of 99.93%. DBN models based on a  $14 \times 42$  Restricted Boltzmann Machine, trained with a learning rate of 0.001 and momentum of 0.5 over 30 pre-training and 100 fine-tuning epochs, achieved approximately 95% accuracy with a classification time of about 0.05 s. Dropout-regularized networks with architectures such as  $14 \times 500 \times 500 \times 500 \times 2$  and dropout probabilities of 0.5 (hidden

**Table 1**  
Summary of prior studies models used for EEG Eye State Classification.

| Study | Model(s)                       | Best Acc. (%) | Validation rigor                               | Explainability     | Robustness considered   | Key Takeaway                                                                 |
|-------|--------------------------------|---------------|------------------------------------------------|--------------------|-------------------------|------------------------------------------------------------------------------|
| [12]  | kNN, MLP                       | 84.06         | Weak (No. mentioned)                           | None               | No                      | Simple baseline study with limited validation and methodological detail      |
| [13]  | LVQ + Bagged Trees (BT)        | 94.31         | Moderate (Stratified 5 fold cross-validation)  | Model-inherent     | No                      | Hybrid clustering-ensemble approach achieves strong accuracy-speed trade-off |
| [10]  | KStar (42 models tested)       | 97.3          | Moderate (Stratified 10 fold cross-validation) | None               | Partial (outliers only) | High accuracy achieved at the expense of substantial computational cost      |
| [14]  | MLP, DBN, Deep Neural Networks | 99.93         | High (80%/20% and 10 fold cross-validation)    | Visualization only | No                      | Deep models achieve very high accuracy with real-time feasibility            |

and 0.3 (input) achieved up to 97.5% accuracy. These deep models demonstrated substantial computational advantages over instance-based methods, achieving at least 30,000 samples per second for MLPs and over 1000 samples per second for dropout networks, compared to approximately 25–27 samples per second for KStar-based classifiers. Despite detailed reporting of architectures and runtime, robustness to missing or corrupted EEG channels was not evaluated, and explainability was limited to t-SNE-based feature visualization (see [Table 1](#)).

Although several prior studies report promising results exceeding those obtained in the present work—such results are often achieved under specific validation strategies and assume fully observed EEG inputs. In particular, leave-one-out cross-validation (LOOCV), while useful for small-sample evaluation, is not directly comparable to k-fold or holdout-based schemes in EEG time-series settings, as it evaluates highly overlapping samples and may yield optimistic performance estimates, especially for single-subject recordings with strong temporal correlation. Recent work, including [15], highlights the importance of adopting validation protocols that more closely reflect real-world deployment conditions. Consequently, improvements in peak accuracy reported under LOOCV or similar settings do not necessarily indicate enhanced robustness or practical reliability. In contrast, this study emphasizes validation consistency across multiple k-fold and fixed-split evaluations, robustness to missing electrode signals, and model interpretability, prioritizing deployment-relevant performance over accuracy optimization on a single benchmark dataset.

### 3. Methodology

In this section, we present the motivation behind our study, described the dataset, outlined the evaluation metrics and algorithms used, and provided an overview of the implementation workflow leading to the final results.

#### 3.1. Motivation

Existing studies focus primarily on accuracy, often overlooking how specific EEG electrodes contribute to or are influenced by eye state changes. This lack of interpretability limits clinical trust and prevents insights into underlying neurophysiological mechanisms. Understanding the role of individual electrodes is essential for validating predictions and aligning model behavior with brain function. EEG signals are also prone to missing values due to sensor detachment, motion

artifacts, and poor contact—making incomplete data a common issue in real-world scenarios. Yet, most models assume clean, complete input or rely on imputation, which can degrade performance and introduce bias. To date, To the best of our knowledge, existing studies on the UCI EEG Eye State dataset primarily assume complete EEG recordings, as they rely on classifiers that require fully observed inputs, and therefore do not explicitly report systematic evaluations of eye state classification performance under controlled and varying levels of missing EEG data.

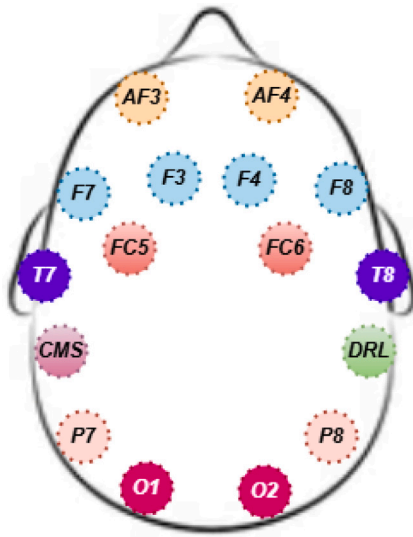
In addressing these signal acquisition challenges, we adopt XGBoost not only for its predictive performance but also because its native handling of missing values presents a principled alternative to conventional imputation-based preprocessing, which can distort signal statistics. This approach allows us to work directly with the raw, incomplete signal recordings, preserving their inherent structure. Furthermore, to advance beyond mere classification and toward meaningful signal characterization, we leverage the model's structure for feature importance analysis, augmented with SHAP and LIME to provide both global and local explanations of electrode influence. This framework allows us to identify which electrode signals — and thus which putative neurophysiological sources — are most discriminative, linking model decisions back to the underlying biosignal origins.

#### 3.2. Data

In our experiments, we used the UCI EEG Eye State dataset [16], which was gathered during a single, continuous EEG recording using an Emotiv EEG Neuroheadset over 117 s. The data captured signals from 14 channels corresponding to electrodes shown in [Fig. 1](#).

The dataset provides the instantaneous voltage measurement from each of the 14 electrodes at 128 Hz, resulting in a time series of spatial potential configurations. Each data point thus represents a snapshot of the full scalp potential field at a given moment, labeled with the corresponding eye state. Consequently, our analysis investigates the discriminative information contained within this spatial representation of the EEG signal, without incorporating derived temporal or spectral features to isolate the contribution of the spatial distribution of scalp potentials to eye state classification—a foundational signal processing question for this specific BCI paradigm.

Each measurement was labeled to indicate the subject's eye state, with '1' representing closed eyes and '0' representing open eyes. In total, the dataset comprised 14,980 samples, all ordered chronologically from the beginning of the recording. Unlike many EEG studies that rely on segmenting continuous recordings into fixed-length windows, the



**Fig. 1.** Placement of electrodes in the Emotiv EEG headset, showing selected frontal (AF3, AF4, F3, F4, F7, F8), temporal (T7, T8), parietal (P7, P8), and occipital (O1, O2) sites. Two additional electrodes are included for referencing and noise reduction: the Common Mode Sense (CMS) electrode, which acts as the active reference channel, and the Driven Right Leg (DRL) electrode, which serves as the ground and helps suppress common-mode electrical interference.

UCI EEG Eye State dataset provided each row as a labeled instance of either eyes open or eyes closed. Therefore, no additional segmentation was applied in our analysis, and the data was used in its provided form. Because the dataset contains single-sample values from 14 electrodes rather than continuous raw waveforms, it is not possible to compute frequency-domain features such as alpha, beta, theta, or delta power. Accordingly, our analysis is restricted to electrode-level values only. Because the dataset originates from a single subject and a short recording session, its generalizability across individuals and recording conditions might be limited; we revisit this issue in the limitations section.

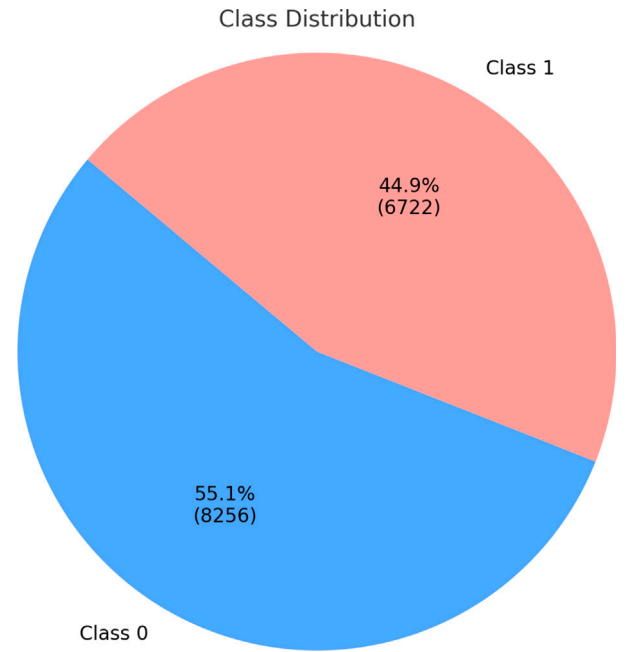
The class imbalance test shown in Fig. 2 demonstrated that the near-balanced distribution supported fair evaluation of model performance without the need for oversampling or downsampling.

### 3.3. Evaluation metrics

To quantify the efficacy of the spatial voltage representation for discriminating eye state, we employed the following standard classification metrics. Accuracy measures the overall proportion of correct predictions out of all predictions (Eq. (A.1)). Precision quantifies the proportion of true positives among all predicted positives (Eq. (A.2)), while Recall — also referred to as Sensitivity or True Positive Rate (TPR) — evaluates the proportion of actual positives correctly identified (Eq. (A.3)). The F1 Score is the harmonic mean of Precision and Recall, providing a balanced assessment particularly valuable for imbalanced datasets (Eq. (A.4)). Finally, AUC represents the model's ability to discriminate between classes across all thresholds (Eq. (A.5)).

### 3.4. XGBoost classifier

In our work, we relied on XGBoost, a gradient boosting algorithm that has become a go-to choice for many classification tasks [17], as a nonlinear, discriminative model for the spatial EEG voltage representation. Its foundation in decision trees makes it inherently suited to partition the high-dimensional, correlated, and non-stationary feature space formed by the 14 simultaneous electrode signals. Crucially, from a signal preprocessing perspective, XGBoost's native handling of missing values provides a principled alternative to conventional imputation



**Fig. 2.** Class distribution in the EEG dataset. Class 0 (eyes open) contains 8256 instances (55.1%), while Class 1 (eyes closed) contains 6722 instances (44.9%). Although there is a mild imbalance (ratio = 1.2:1), it is not large enough to require resampling techniques. This distribution is relatively balanced for training and evaluating classification models without introducing bias.

techniques [18], allowing the model to learn optimal decisions from incomplete recordings without introducing the potential bias of filled values—a critical advantage for real-world EEG data prone to artifacts and sensor dropout [18].

The key hyperparameters used in our XGBoost model, along with their default values, are summarized in the following Table 2.

### 3.5. Grid search

To ensure that the XGBoost classifier was properly calibrated to the statistical characteristics of the spatial EEG voltage signals, we employed a grid search for hyperparameter optimization [19]. This exhaustive search across a defined parameter space serves not merely to maximize classification scores, but to mitigate overfitting to noise or underfitting of the underlying signal structure [19], both critical concerns when working with high-dimensional, noisy biosignals. A well-tuned model provides a more reliable and valid estimate of the inherent separability of the two eye state conditions based on the raw voltage representation, which is the core signal processing question under investigation.

### 3.6. Explainability methods

To move beyond aggregate classification performance and characterize which aspects of the spatial EEG signal were most discriminative, we analyzed the trained model using XGBoost's built-in feature importance alongside SHAP and LIME. These methods quantify the contribution of each electrode to the model's decisions, both globally across the dataset and locally for individual instances. This analysis transforms the classifier into an analytical tool for source attribution, providing electrode-level insights that can be mapped back to neurophysiological origins and validating whether the model's discrimination aligns with known physiological principles of eye state modulation in EEG.



**Table 2**  
Common XGBoost Hyperparameters and Their Default Values.

| Parameter        | Description                                                                     | Default value   |
|------------------|---------------------------------------------------------------------------------|-----------------|
| booster          | Specifies the type of model used: tree-based, linear, or DART.                  | gbtree          |
| eta              | Step size shrinkage to prevent overfitting.                                     | 0.3             |
| gamma            | Minimum loss reduction required to make a split. Encourages conservative trees. | 0               |
| max_depth        | Maximum depth of each tree to control complexity.                               | 6               |
| min_child_weight | Minimum sum of instance weights (hessian) needed in a child.                    | 1               |
| subsample        | Fraction of training instances used per tree to prevent overfitting.            | 1.0             |
| colsample_bytree | Fraction of features used per tree.                                             | 1.0             |
| lambda           | $L_2$ regularization term on leaf weights.                                      | 1               |
| alpha            | $L_1$ regularization term on leaf weights.                                      | 0               |
| scale_pos_weight | Balances positive and negative class weights for imbalanced data.               | 1               |
| objective        | Specifies the learning task and corresponding loss function.                    | binary logistic |

### 3.6.1. Feature importance and gain criterion in XGBoost

XGBoost provides a signal-centric measure of electrode relevance by quantifying feature importance through the *Gain*, defined as the reduction in the objective (loss) function when an electrode's signal is used to split the data [17]. The Gain is derived from the second-order Taylor expansion of the objective (Eq. (A.6)), where aggregated gradient ( $G$ ) and Hessian ( $H$ ) statistics (Eq. (A.7)) determine the optimal leaf weight (Eq. (A.8)) and split quality (Eq. (A.9)), with the improvement quantified using the Gain formulation (Eq. (A.10)). In this work, each feature corresponds to the time-domain voltage of a single EEG electrode. Electrode importance is obtained by summing the Gain across all splits where the electrode is used (Eq. (A.11)), yielding a global ranking that reflects how effectively the spatial and amplitude information of each channel discriminates between eyes-open and eyes-closed states. This interpretation enables the ensemble model to serve as a tool for electrode-level signal-source assessment, highlighting cortical regions most informative for the physiological task.

### 3.6.2. Local interpretable model-agnostic explanations

We employ LIME to perform a sample-wise sensitivity analysis of the classifier's decisions for individual EEG moments, providing localized explanations that link model outputs to specific spatial voltage patterns [20]. The method operates by generating perturbed versions of a given multi-channel voltage sample and weighting them by their proximity to the original signal. A simple, interpretable surrogate model (e.g., a weighted linear regression) is then fitted to approximate the complex model's behavior in this local region. This process is formalized as an optimization problem (Eq. (A.12)), where the loss (Eq. (A.13)) quantifies how well the surrogate approximates the original model, and a penalty term (Eq. (A.14)) enforces simplicity. The solution yields a set of feature weights (Eq. (A.15)) that quantify the local influence of each electrode's voltage measurement on the prediction, thereby translating the classifier's decision into an interpretable map of spatial-importance for that specific EEG instance.

### 3.6.3. Shapley additive explanations

We employ SHAP as a principled signal-attribution framework to decompose XGBoost predictions for a given EEG sample into additive

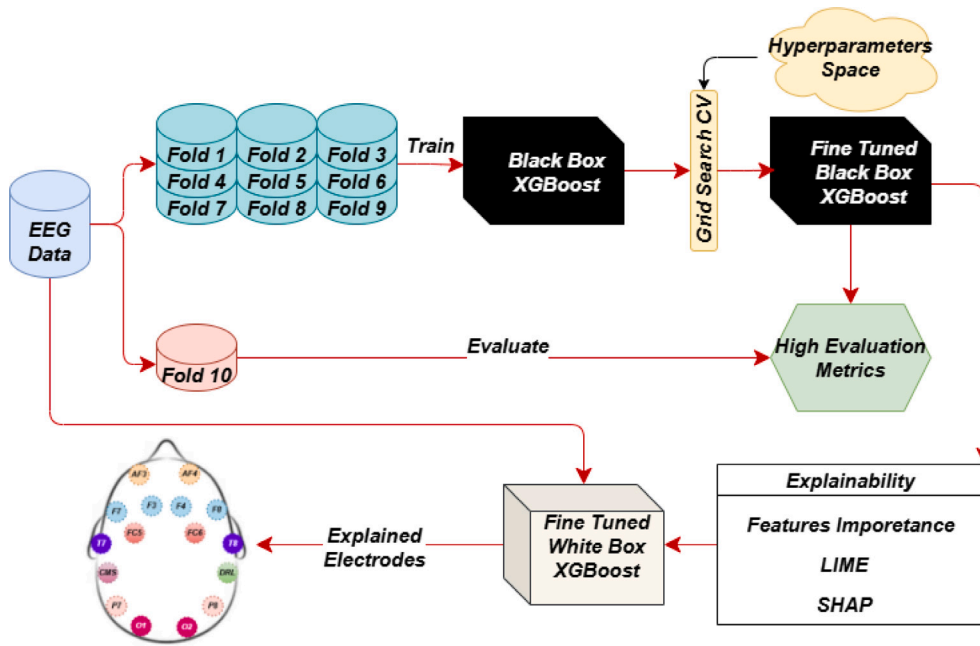
contributions from each electrode. Grounded in cooperative game theory, SHAP assigns each electrode a value (Eq. (A.17)) corresponding to its average marginal contribution to the prediction across all feature combinations, yielding a locally additive explanation (Eq. (A.16)) that quantifies how individual voltage measurements shift the output from a baseline. By averaging the absolute SHAP values across all samples (Eq. (A.18)), a global ranking of electrode importance is obtained. This links instance-level voltage patterns to consistently influential electrode signals, providing a statistically grounded characterization of the spatial information content of the EEG signal for eye state discrimination [21].

### 3.7. Workflow

Our workflow, illustrated in Fig. 3, is designed to rigorously evaluate the discriminative capacity of the spatial EEG voltage representation and to characterize the signal sources that contribute to this discrimination. We first established a performance baseline by comparing our chosen model against several other classifiers, using statistical significance testing to ensure any observed differences were robust. To obtain a reliable estimate of performance inherent to the signal representation, we employed k-fold cross-validation. We then conducted a model calibration phase using a grid search to optimize hyperparameters, which is critical for preventing overfitting to noise and underfitting of the underlying signal structure in the high-dimensional EEG data. The robustness of this calibrated model was further tested under varied data-partitioning schemes (different k-fold configurations and a standard train-test split).

To assess the practical viability of the approach given common signal acquisition issues, we introduced synthetic missing values into the feature matrix under a Missing Completely at Random (MCAR) assumption, testing the model's resilience to the kind of incomplete data typical of real-world EEG recordings.

Finally, we applied explainable AI (XAI) techniques — specifically global and local SHAP explanations alongside LIME — to transform the classifier into an analytical tool. This step characterizes the contribution of each electrode signal to the model's decisions, providing both global feature importance rankings and instance-specific explanations that link predictions back to the spatial voltage patterns.



**Fig. 3.** Workflow of the proposed EEG signal analysis and classification framework. The process begins with the raw spatial voltage representation (14-channel data). Following a comparative model assessment, the core pipeline uses stratified cross-validation to train and calibrate (via grid search) the XGBoost classifier, thereby evaluating the signal's inherent separability. The framework then tests robustness to simulated acquisition artifacts (missing data) and concludes with explainability methods that analyze and attribute the discriminative output to specific electrode signals, moving from a predictive model to an interpretable signal characterization tool.

### 3.8. Experimental environment

All experiments were executed on a workstation equipped with an AMD Ryzen 5 3600 CPU (6 cores, 12 threads), 32 GB DDR4 RAM (CL16, 3333 MHz), no GPU acceleration was neither needed or used. The experimental workflow was implemented in the PyCharm integrated development environment using Python 3.10. The machine learning experiments were conducted with scikit-learn (v1.7.1) and XGBoost (v3.1.1).

## 4. Implementation

This section outlines the computational framework and validation protocols used to ensure a rigorous and reproducible analysis of the EEG signal's discriminative capacity. All stochastic processes, including data partitioning, model training, and explanation generation, were controlled using a fixed random seed (`random_state = 42`).

### 4.1. Data split

To obtain a robust and unbiased estimate of the discriminative performance of the spatial EEG voltage representation, we employed 10-fold cross-validation. This approach evaluates generalization by iteratively training and testing on disjoint data subsets. Stratified sampling was used within each fold to preserve the original eye state class proportions, ensuring that performance estimates reflect intrinsic signal separability rather than class imbalance effects.

### 4.2. Baseline model

To justify our choice of XGBoost, we carried out two complementary experiments: comparative performance evaluation against baseline models and significance testing to confirm the robustness of its superiority.

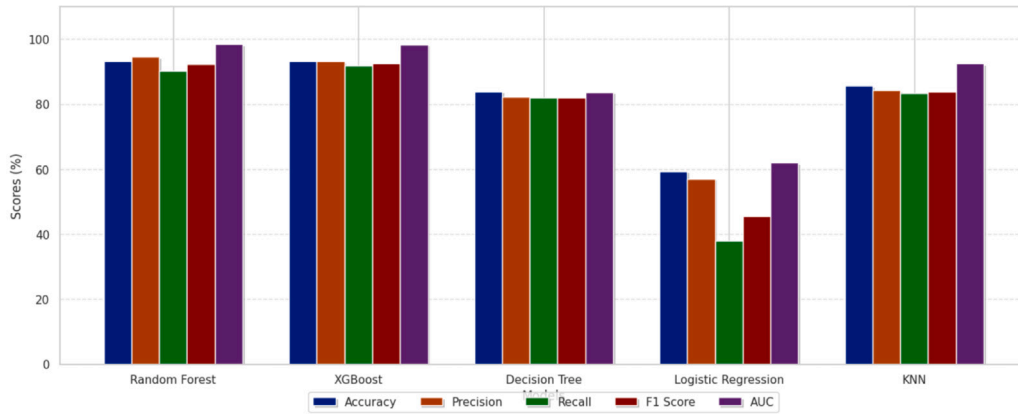
#### 4.2.1. Comparative model evaluation

To position XGBoost as the go-to signal analysis tool and verify that simpler models could not extract comparable information from the spatial EEG voltage representation, we performed a comparative evaluation against baseline classifiers, including Decision Tree (DT), kNN, Logistic Regression, and Random Forest (RF). All models were evaluated using the same 10-fold cross-validation protocol (Section 4.1) to ensure a fair assessment, with data scaling applied only to Logistic Regression and kNN, while tree-based models operated on raw voltage values. In addition to mean performance scores, 95% confidence intervals were computed from fold-level results to assess statistical reliability. As shown in Fig. 4, ensemble-based methods (Random Forest and XGBoost) consistently outperformed linear and distance-based approaches, supporting the use of a nonlinear ensemble model to capture the complex, high-dimensional structure of the multi-channel EEG signal.

Both XGBoost and RF exhibited strong and statistically comparable performance, as evidenced by their overlapping confidence intervals (Table 3). Although their aggregate results were similar, XGBoost was selected for subsequent analysis due to signal-relevant advantages, including native handling of missing values and an efficient implementation that supports the extensive calibration and explainability procedures central to our signal characterization framework. In addition, the slightly higher recall achieved by XGBoost aligns with the practical objective of reliably identifying the eyes-closed state.

#### 4.2.2. Significance testing and computational analysis

To statistically validate the suitability of XGBoost as a signal analysis tool, we conducted pairwise significance tests against Random Forest (RF) using both paired *t*-tests and Wilcoxon signed-rank tests. As reported in Table 4, differences in overall discriminative performance, measured by accuracy and AUC, were not statistically significant ( $p > 0.3$ ). Although significant differences were observed in precision and recall ( $p < 0.01$ ), with RF favoring precision and XGBoost favoring recall, the composite F1 score — which balances both measures — also showed no significant difference. These findings confirm that both models act as statistically equivalent core discriminators of the EEG signal



**Fig. 4.** Comparative performance of five classifiers in evaluating the separability of the spatial EEG voltage representation. Results, averaged over 10-fold cross-validation, show that tree-ensemble methods (Random Forest and XGBoost) achieve superior and statistically robust discrimination compared to linear (Logistic Regression) and distance-based (kNN) models. This supports the use of a nonlinear, ensemble-based approach as a sufficiently powerful signal analysis tool for capturing the discriminative information present in the multi-channel EEG data for eye state classification.

**Table 3**

Mean evaluation metrics of classification models with 95% confidence intervals (10-Fold Cross-Validation).

| Model   | Accuracy (%) | Precision (%) | Recall (%)   | F1 Score (%) | AUC (%)      |
|---------|--------------|---------------|--------------|--------------|--------------|
| RF      | 93.29 ± 1.10 | 94.61 ± 0.95  | 90.20 ± 1.50 | 92.35 ± 1.20 | 98.39 ± 0.40 |
| XGBoost | 93.30 ± 1.05 | 93.25 ± 1.10  | 91.73 ± 1.35 | 92.48 ± 1.15 | 98.34 ± 0.42 |
| DT      | 83.86 ± 2.20 | 82.15 ± 2.40  | 81.87 ± 2.30 | 82.00 ± 2.25 | 83.67 ± 2.10 |
| LR      | 59.37 ± 3.80 | 57.06 ± 3.95  | 37.88 ± 4.50 | 45.52 ± 4.20 | 61.93 ± 3.70 |
| kNN     | 85.53 ± 1.95 | 84.31 ± 2.00  | 83.27 ± 2.10 | 83.78 ± 2.05 | 92.56 ± 1.50 |

**Table 4**

Pairwise significance tests between XGBoost and RF (10-fold cross-validation). Stat = test statistic,  $p$  =  $p$ -value. Significant results ( $p < 0.05$ ) are in bold.

| Metric    | t-test (Stat) | t-test ( $p$ ) | Wilcoxon (Stat) | Wilcoxon ( $p$ ) |
|-----------|---------------|----------------|-----------------|------------------|
| Accuracy  | 0.053         | 0.959          | 24.5            | 0.846            |
| Precision | -3.384        | <b>0.008</b>   | 3.0             | <b>0.010</b>     |
| Recall    | 3.410         | <b>0.008</b>   | 3.0             | <b>0.010</b>     |
| F1 Score  | 0.466         | 0.652          | 21.0            | 0.557            |
| AUC       | -0.887        | 0.398          | 19.0            | 0.432            |

**Table 5**

Friedman and Nemenyi post-hoc tests across all models. Significant differences ( $p < 0.05$ ) highlight superior groups.

| Metric    | Friedman stat ( $p$ ) | Significant differences (Nemenyi)      |
|-----------|-----------------------|----------------------------------------|
| Accuracy  | 34.64 ( $p < 0.001$ ) | XGBoost, RF > DT, LR, kNN              |
| Precision | 36.08 ( $p < 0.001$ ) | RF > DT, LR, kNN; XGBoost > LR         |
| Recall    | 34.59 ( $p < 0.001$ ) | XGBoost > LR, DT; RF > LR              |
| F1 Score  | 34.64 ( $p < 0.001$ ) | XGBoost, RF > DT, LR, kNN              |
| AUC       | 37.04 ( $p < 0.001$ ) | XGBoost, RF > DT, LR; kNN intermediate |

representation. Consequently, the selection of XGBoost is motivated by its methodological advantages within the proposed pipeline, such as native handling of missing data, rather than by a decisive performance advantage.

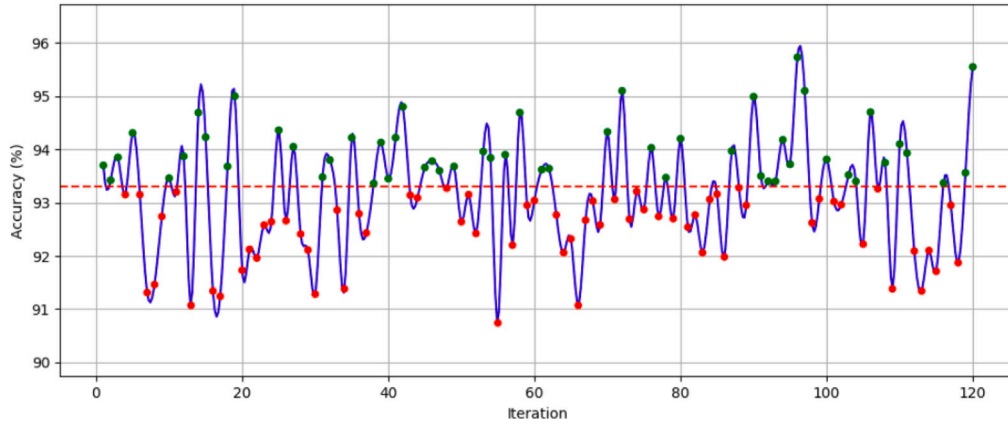
We further applied the Friedman test across all five models, followed by the Nemenyi post-hoc test, to identify global rank differences. The Friedman test returned highly significant results ( $p < 0.001$ ) across all metrics, confirming that model performances were not equivalent. As summarized in Table 5, post-hoc analysis revealed that XGBoost and RF consistently formed the top-performing group, significantly outperformed DT, LR, and kNN across most metrics.

Based on the statistical equivalence in core discriminative metrics (accuracy, AUC, F1) established by the significance tests (Tables 4 and 5) and the performance profiles in Table 3, XGBoost was selected for the subsequent analysis. This choice was not predicated on a decisive performance superiority but on its signal-processing-aligned characteristics: its native handling of missing data provides a principled alternative to imputation, and its structure seamlessly integrates

with the explainability techniques required for electrode-level signal characterization.

#### 4.3. Hyperparameter tuning

To fine-tune the XGBoost model, we employed Grid Search over a limited hyperparameter space, focusing on hyperparameters that are known to strongly influence tree complexity, model generalization, and robustness. Specifically, `max_depth` was tuned because deeper trees can capture complex nonlinear feature interactions but also risk overfitting, making it critical to identify an optimal balance. We explored values of [6, 8, 9, 12]. Similarly, `n_estimators` (the number of boosting rounds) directly controls the ensemble size: too few iterations may lead to underfitting, while very large numbers yield diminishing returns and longer training times. We therefore examined values of [100, 200, 300, 400, 500]. Finally, `subsample` was included to regulate randomness during tree construction, with values of [0.5, 0.6, 0.7,



**Fig. 5.** Accuracy of the XGBoost model across 120 Grid Search iterations, each corresponding to a unique hyperparameter combination. Green markers indicate peak accuracies, red markers indicate dips, and the dashed red line shows the baseline accuracy. While fluctuations were observed across iterations, the model demonstrated steady improvements with multiple peaks above 95%. The best result was achieved at iteration 96, yielding an accuracy of 95.73% using  $\text{max\_depth} = 8$ ,  $\text{n\_estimators} = 500$ , and  $\text{subsample} = 0.8$ .

**Table 6**

Performance of the fine-tuned XGBoost model across different validation strategies.

| Metric        | 3-Folds | 5-Folds | 7-Folds | 10-Folds | split (80/20) |
|---------------|---------|---------|---------|----------|---------------|
| Accuracy (%)  | 94.68   | 95.29   | 95.25   | 95.73    | 96.23         |
| Precision (%) | 94.68   | 95.30   | 95.35   | 95.69    | 96.60         |
| Recall (%)    | 93.49   | 94.14   | 94.02   | 94.76    | 94.94         |
| F1 Score (%)  | 94.04   | 94.72   | 94.68   | 95.22    | 95.76         |
| AUC (%)       | 98.78   | 99.00   | 99.09   | 99.13    | 99.30         |

0.8, 0.9, 1.0] allowing us to test the trade-off between regularization and predictive stability.

Alternatives such as Random Search or Bayesian Optimization are often advantageous in larger or high-dimensional search spaces, but given our focused hyperparameter set and dataset size, Grid Search was a practical and rigorous choice to calibrate the model to the statistical properties of the EEG signal representation while maintaining full experimental transparency and reproducibility.

The tuning process was carried out using the same 10-fold stratified cross-validation strategy employed for training and evaluating the base model, ensuring consistency and reliable comparison.

This optimized setting was fixed and subsequently used in all further evaluations, ensuring consistent high performance.

#### 4.4. Generalization testing across validation strategies

To obtain a reliable and partition-independent estimate of the EEG signal's discriminative capacity, the fine-tuned XGBoost model (using optimal hyperparameters from Section 4.3) was evaluated under multiple validation strategies, including 3-fold, 5-fold, 7-fold, and 10-fold cross-validation, as well as a conventional 80/20 train-test split. This strategy reduces performance variability arising from the randomness of a single data partition, which is unrelated to the intrinsic information content of the signal. By assessing performance across multiple partitioning schemes, we ensure that the estimated separability of the spatial voltage representation is robust and not dependent on a favorable data split. As summarized in Table 6, the model achieved consistently high performance across all strategies; in particular, the 10-fold cross-validation yielded an accuracy of 95.73%, precision of 95.69%, recall of 94.76%, an F1 score of 95.22%, and an AUC of 99.13%. The low variability across validation protocols indicates that the discriminative performance of the spatial voltage representation is stable, supporting the conclusion that it contains substantial information for eye state discrimination.

Table 6 presents the performance of the fine-tuned XGBoost model evaluated under various data splitting strategies. This evaluation was conducted to assess the model's robustness across different testing conditions and to minimize the potential bias introduced by any single split configuration. The results demonstrated consistently strong performance, indicating that the model was both stable and reliable across all validation scenarios.

#### 4.5. Error analysis

To better understand model limitations beyond aggregate performance metrics, we conducted an error analysis using confusion matrices, calibration analysis, and hardness analysis to examine misclassification patterns, confidence reliability, and consistently difficult samples.

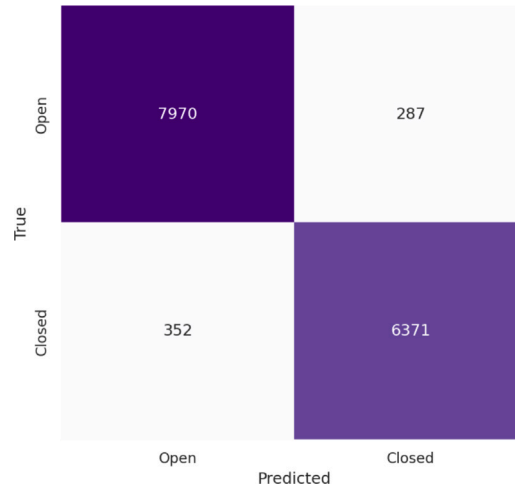
##### 4.5.1. Confusion matrix

The confusion matrix was computed to analyze the symmetry and type of misclassifications between the two eye states, providing insight into whether the model (and thus the signal representation) had a systematic bias toward one physiological condition. It was generated by aggregating predictions from the tuned XGBoost model across all folds of the stratified 10-fold cross-validation (Section 4.1), where rows represent true classes and columns correspond to predicted classes.

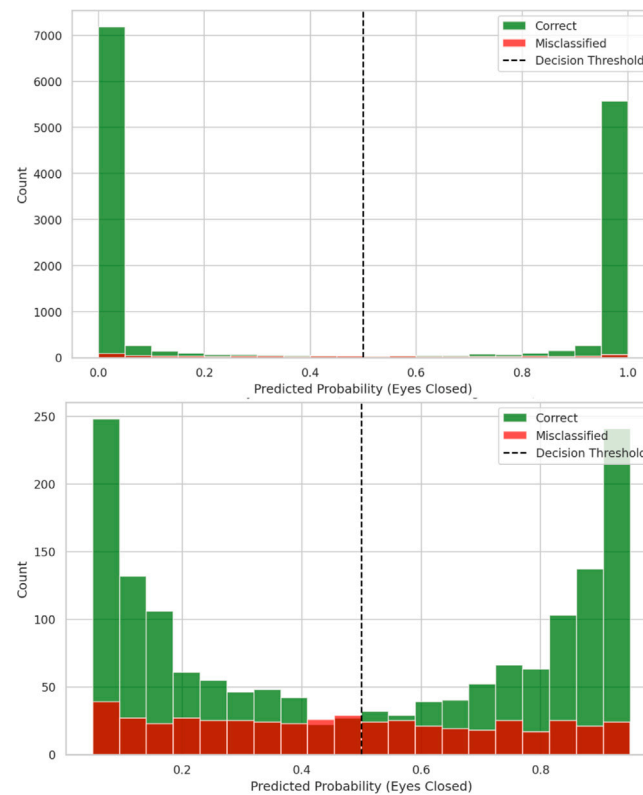
##### 4.5.2. Ambiguity check

To assess whether the model's prediction confidence reflects inherent ambiguity in the EEG signal, we analyzed the distribution of predicted probabilities for the 'eyes closed' class. Probabilities were generated by the tuned XGBoost model using a decision threshold of 0.5. By visualizing separate histograms for correctly classified and misclassified samples, we can examine if errors are concentrated in low-confidence regions, indicating that the model is uncertain — and the signal may be ambiguous — where mistakes occur.





**Fig. 6.** Confusion matrix for the tuned XGBoost classifier, aggregated over 10-fold cross-validation. The matrix illustrates the distribution of discriminative errors between the two physiological states. Of the 8257 eyes-open samples, 287 (3.5%) were misclassified as eyes closed, while of the 6723 eyes-closed samples, 352 (5.2%) were misclassified as eyes open. This asymmetry in error rates suggests that the spatial voltage patterns associated with the eyes-closed condition are marginally more confusable with the eyes-open state, or that the signal exhibits greater ambiguity during eyes-closed periods within this representation.

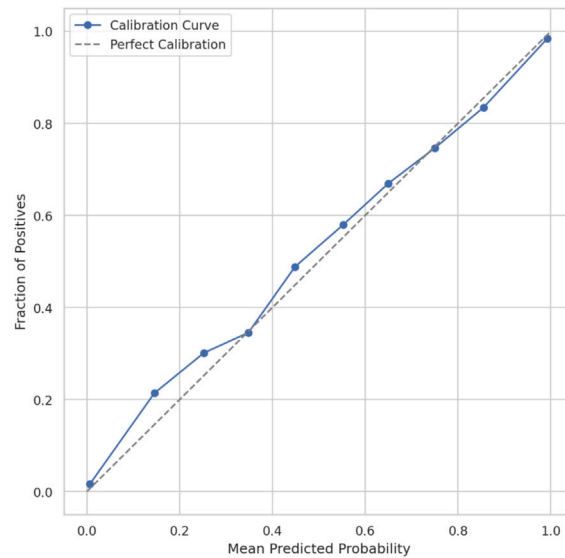


**Fig. 7.** Probability distributions for correctly classified (green) and misclassified (red) EEG samples. The top panel displays the full probability range, showing that most predictions cluster confidently near 0 (eyes open) and 1 (eyes closed). The bottom panel provides a zoomed view of the intermediate region (0.2–0.8), where misclassifications are concentrated near the 0.5 decision threshold. This pattern indicates that errors primarily occur when the instantaneous spatial voltage configuration does not strongly correspond to either canonical eye state, reflecting regions of the signal space where class separability based on amplitude information alone is limited. In contrast, predictions at probability extremes correspond to highly discriminative voltage patterns and are therefore highly reliable.

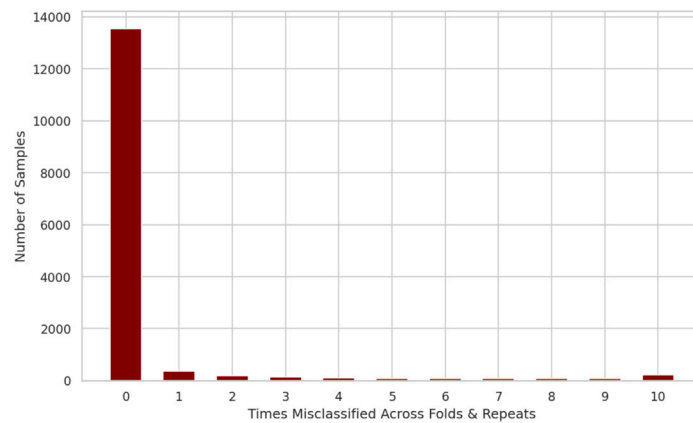
#### 4.5.3. Calibration curve

To assess whether the predicted probabilities accurately reflect the true likelihood of the eyes-closed state given an observed EEG

voltage pattern, we conducted a calibration analysis. Predicted probabilities from the tuned XGBoost model were aggregated across all cross-validation folds and partitioned into 10 uniformly spaced bins over the interval [0, 1]. For each bin, the mean predicted probability was



**Fig. 8.** Calibration curve of the tuned XGBoost classifier, relating predicted probabilities to the observed fraction of positive cases. The close alignment of the blue curve with the dashed diagonal indicates strong calibration. This means the model's predicted probability for an EEG sample can be interpreted as a trustworthy estimate of the likelihood that its spatial voltage pattern corresponds to the eyes-closed state, rather than merely a rank score for classification. Such reliability is crucial for applications where confidence in the physiological state assessment is required.



**Fig. 9.** Hardness analysis of EEG samples based on repeated stratified 10-fold cross-validation. The histogram shows the frequency of misclassification across 100 evaluation rounds. The distribution reveals that the vast majority of spatial voltage patterns are highly discriminative, with over 13,500 samples never misclassified. A small subset was misclassified occasionally, and only 227 samples were consistently misclassified in all repetitions. These persistently hard samples represent EEG patterns where the spatial voltage configuration is inherently ambiguous or lies very close to the class boundary in the feature space, defining the practical limits of separability for the chosen signal representation.

compared with the observed fraction of eyes-closed samples, yielding the calibration curve. A diagonal reference line was included to indicate perfect calibration.

#### 4.5.4. Hardness analysis

To identify inherently ambiguous EEG patterns that challenge the discriminability of the spatial voltage representation, we performed a hardness analysis that quantifies the difficulty of individual samples based on their frequency of misclassification. The analysis was conducted using the tuned XGBoost model with repeated stratified 10-fold cross-validation over 10 repetitions, preserving class proportions in each fold. During each fold, a sample's hardness score was incremented whenever its predicted label differed from the ground truth. After completing all folds and repetitions, misclassification counts were aggregated per sample. In this configuration, each sample could be

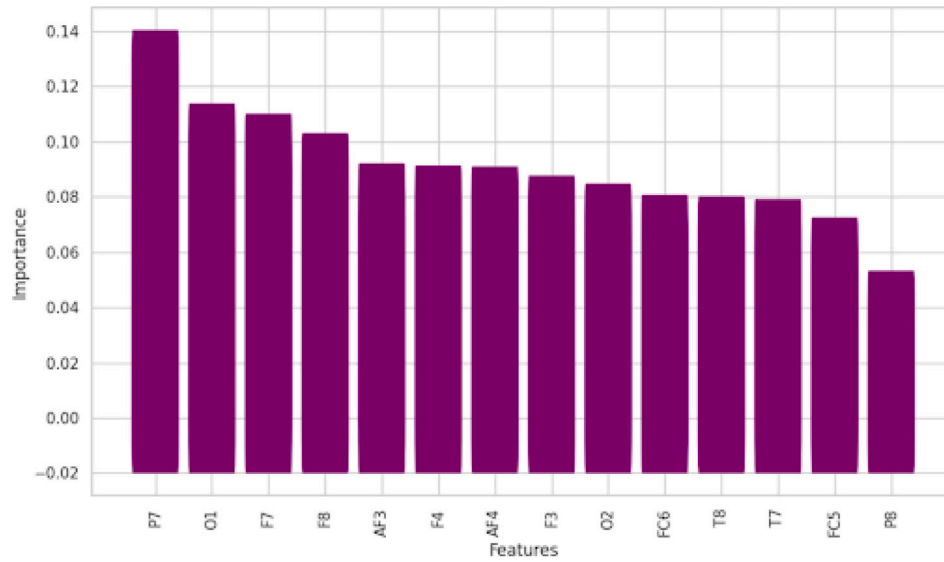
misclassified up to 100 times (10 folds  $\times$  10 repetitions), with higher counts indicating greater ambiguity or difficulty.

#### 4.6. Explainability and interpretability

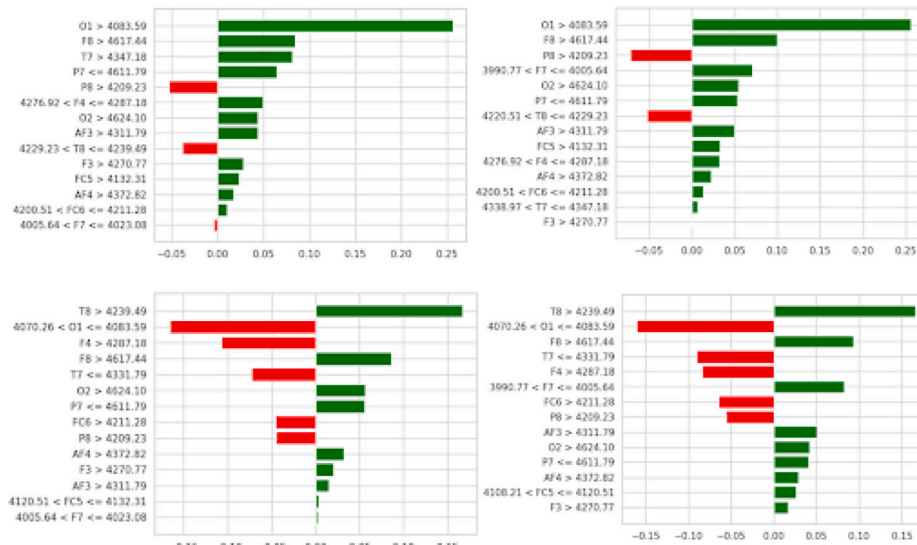
In this section, we apply three XAI approaches — feature importance, LIME, and SHAP — to analyze which electrode signals were most influential for the eye state discrimination, thereby characterizing the spatial origins of the discriminative information in the EEG recording.

##### 4.6.1. Feature importance

The first step in our signal characterization was to rank the contribution of each electrode using the tuned XGBoost model's internal feature importance, calculated via the Gain criterion as defined in Section 3.6.1.



**Fig. 10.** Feature importance ranking of EEG electrodes for eye state discrimination. Electrodes P7, O1, F7, and F8, located over parietal, occipital, and frontal regions, emerged as the most discriminative. This finding indicates that the spatial voltage patterns most informative for eye state classification originate primarily from posterior and lateral scalp areas, which are neurophysiologically associated with visual processing and eye-movement activity.



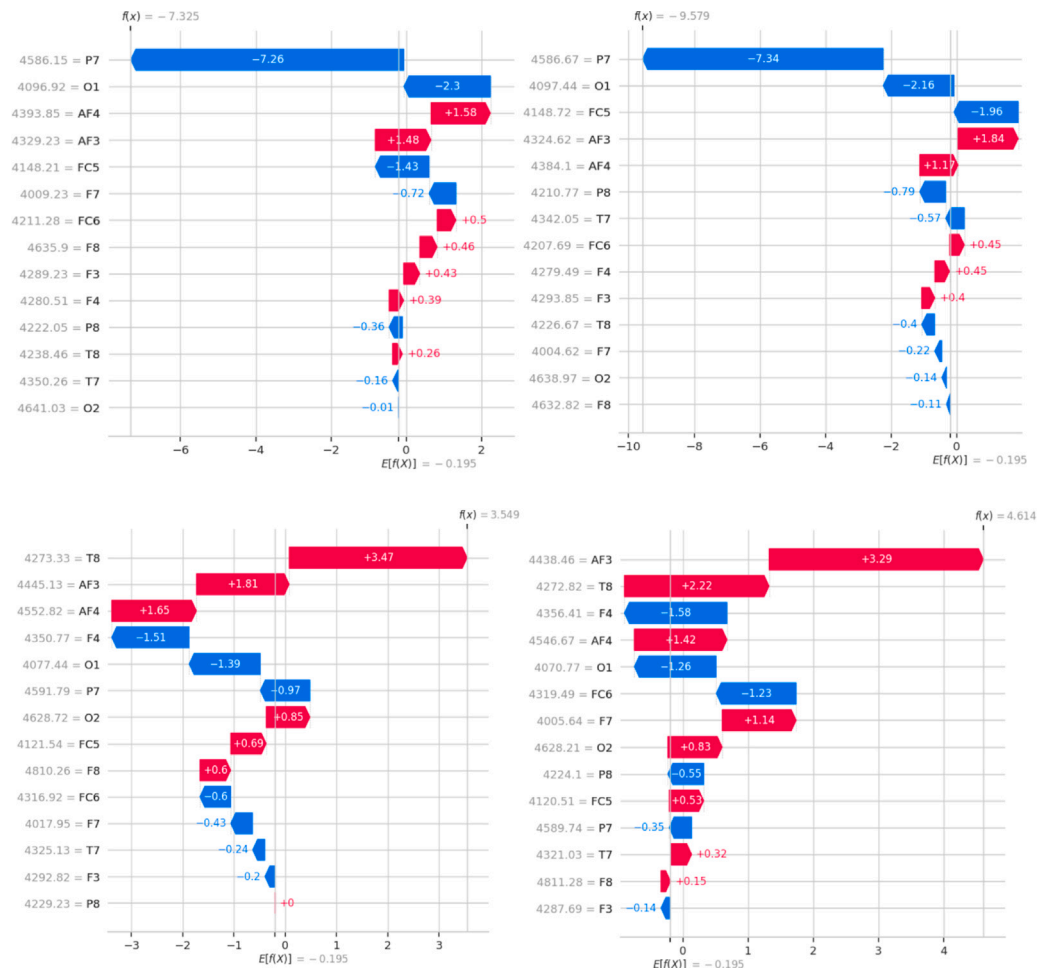
**Fig. 11.** LIME feature importance plots for four EEG instances, with two examples from Class 0 (eyes open) shown in the top row and two from Class 1 (eyes closed) shown in the bottom row. Green bars denote positive contributions toward the predicted class, while red bars indicate negative contributions. The results show that different subsets of electrodes can drive the same classification outcome, highlighting variability in the spatial voltage patterns associated with each physiological state. This instance-level analysis complements the global feature importance by demonstrating how discriminative spatial configurations can change across time in the EEG signal.

#### 4.6.2. Local LIME

To examine electrode-level contributions driving individual classification decisions, we applied LIME as described in Section 3.6.2. Continuous features were discretized, 500 perturbed samples were generated per instance, and a kernel width of 0.75 was used to weight perturbations by proximity. Four representative samples — two from the eyes-open class (Class 0) and two from the eyes-closed class (Class 1) — were selected for local analysis. Explanations included all EEG features, retaining only contributions with normalized magnitude  $|contribution| \geq 0.01$  to suppress negligible effects while preserving meaningful electrode attributions.

#### 4.6.3. Local SHAP

To quantify electrode-level contributions to individual predictions, we applied local SHAP analysis using the tuned XGBoost model. Four representative instances were selected to preserve class balance—two from the eyes-open class (Class 0) and two from the eyes-closed class (Class 1). For each instance, SHAP waterfall plots were generated to report the base value and the contribution of all 14 electrodes to the prediction. To improve interpretability and suppress noise-driven effects, only features with normalized contributions of  $|contribution| \geq 0.01$  were retained in the visualizations and subsequent analysis.



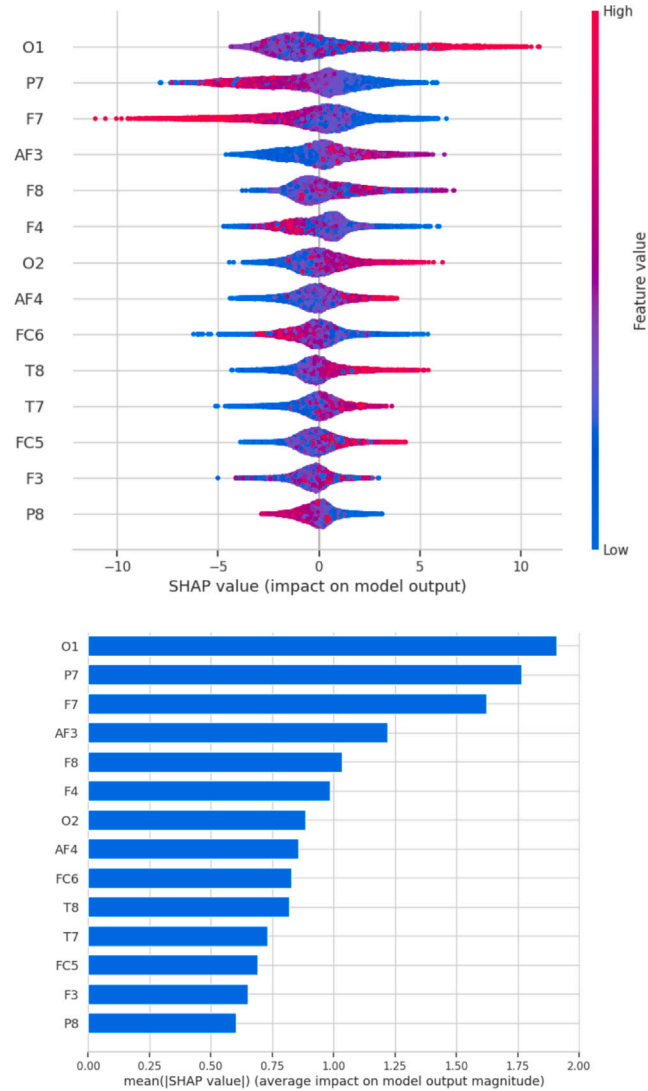
**Fig. 12.** Local SHAP waterfall plots for four EEG instances, including two eyes-open samples (Class 0, top) and two eyes-closed samples (Class 1, bottom). Positive SHAP values (red) shift predictions toward the eyes-closed state, while negative values (blue) favor the eyes-open state. Across samples, posterior electrodes (P7, O1) show strong negative contributions for eyes-open predictions, whereas frontal electrodes (AF3, AF4) contribute positively to eyes-closed predictions. This consistent pattern indicates that the model captures a neurophysiological signature characterized by reduced posterior and increased frontal activity in the eyes-closed state.

#### 4.6.4. Global SHAP

To identify electrode signals that consistently influenced model predictions across the entire EEG recording, global SHAP values were computed by averaging the absolute feature attributions over all samples. SHAP values were calculated for the full dataset to ensure stable

estimation of feature importance, yielding a ranking of electrodes based on their overall influence across classes. To emphasize robust contributions, a normalized threshold of  $|contribution| \geq 0.01$  was applied, filtering out minor effects.





**Fig. 13.** Global SHAP explainability results. The top plot shows the SHAP summary distribution, where each point represents an individual sample's feature contribution (red = high feature value, blue = low feature value). This illustrates both the direction and magnitude of each electrode's impact on the model output. The bottom plot ranks electrodes by their mean absolute SHAP values, reflecting overall importance. Together, these plots highlight O1, P7, and F7 as the strongest global predictors: O1 (occipital) and P7 (parietal) support eye state discrimination through visual and attentional signals, while F7 (frontal) reflects executive control and alertness monitoring. Frontal electrodes AF3 and AF4 also contribute meaningfully, while P8 consistently shows the lowest influence.

#### 4.7. Robustness of the tuned model under incomplete data

To assess the practical viability of the proposed approach under a key EEG acquisition challenge, we evaluated robustness to incomplete data caused by artifacts, poor electrode contact, or sensor dropout. Two complementary experiments were conducted: (i) uniform data loss via random removal across all features, and (ii) targeted deletion focusing on top-ranked electrodes identified through explainability analysis. In both cases, missing values were introduced under an MCAR assumption. For each missingness level (1%, 3%, 5%, 7%, 10%, and 20%), feature values were uniformly sampled and replaced with NaN, ensuring independence from class labels and feature distributions.

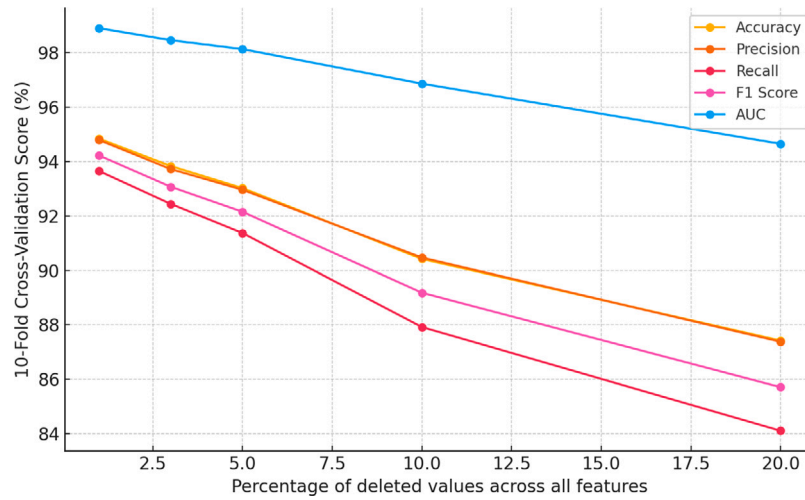
##### 4.7.1. Random deletion

This experiment simulates generalized, non-systematic signal degradation, such as that arising from transient noise or intermittent poor

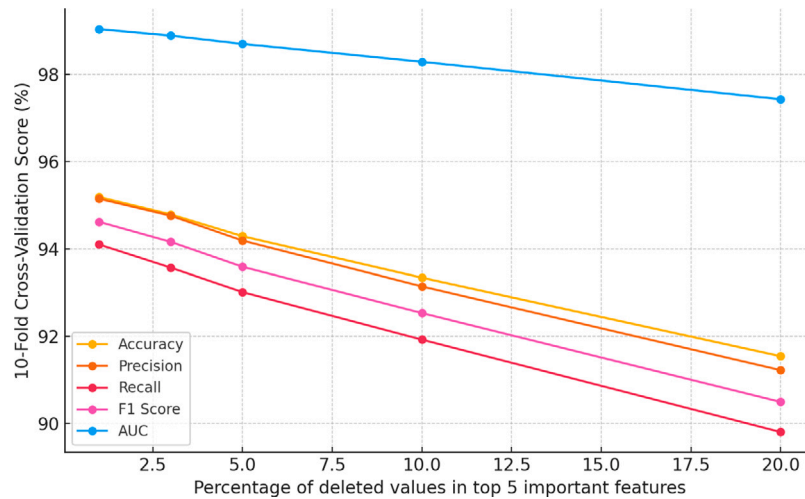
contact across multiple electrodes. Missing values were uniformly introduced across all features. The tuned model was then re-evaluated using the 10-fold cross-validation procedure described in Section 4.1, and performance metrics were recorded to quantify the impact of this diffuse, random data loss.

##### 4.7.2. Targeted deletion

This experiment conducts a sensitivity analysis to assess the criticality of the most discriminative signal sources. Missing values were introduced exclusively into the five highest-ranked electrodes — O1, P7, F7, AF3, and F8 — as identified by the global explainability analysis, while all remaining features were left intact. The tuned model was then evaluated using 10-fold cross-validation, and performance metrics were recorded to quantify the impact of this targeted, feature-specific signal corruption.



**Fig. 14.** Impact of uniformly missing data on model performance. Mean 10-fold cross-validation scores for the tuned XGBoost model under increasing proportions of randomly deleted values across all EEG features are reported. All metrics decline gradually with increasing missingness, with a more pronounced degradation beyond 10%. This behavior suggests that the spatial voltage representation retains substantial discriminative information under moderate diffuse corruption, but performance deteriorates once a critical fraction of the signal is removed, reflecting the distributed nature of multi-channel EEG.



**Fig. 15.** Impact of missing data in the most influential features on model performance. The plot shows mean 10-fold cross-validation scores for the tuned XGBoost model when random values are deleted from the top five EEG features ranked by importance. Accuracy, precision, recall, F1 score, and AUC all decline more steeply compared to deletion across all features, demonstrating the model's higher sensitivity to missing values in key electrodes.

## 5. Discussion

The training phase began with a comparison of XGBoost against several baseline models, including DT, RF, LR, and kNN. Using 10-fold cross-validation with 95% confidence intervals as a consistent evaluation framework, XGBoost emerged as one of the top performers (Table 3). It consistently outperformed DT, LR, and kNN, achieving notably higher accuracy ( $93.30 \pm 1.05\%$  vs.  $83.86 \pm 2.20\%$  for DT, and  $59.37 \pm 3.80\%$  for LR) and F1 score ( $92.48 \pm 1.15\%$  vs.  $82.00 \pm 2.25\%$  for DT, and  $45.52 \pm 4.20\%$  for LR), while also surpassing kNN in both accuracy ( $93.30 \pm 1.05\%$  vs.  $85.53 \pm 1.95\%$ ) and AUC ( $98.34 \pm 0.42\%$  vs.  $92.56 \pm 1.50\%$ ). Its performance was closely aligned with RF, with XGBoost showing slightly higher recall ( $91.73 \pm 1.35\%$  vs.  $90.20 \pm 1.50\%$ ) and F1 score ( $92.48 \pm 1.15\%$  vs.  $92.35 \pm 1.20\%$ ), whereas RF achieved marginally higher precision ( $94.61 \pm 0.95\%$ ). Pairwise statistical tests (Table 4) confirmed that these differences were significant for precision and recall ( $p < 0.01$ ), with RF favoring precision and XGBoost favoring recall. Global tests across all models (Table 5) further showed that XGBoost and RF consistently formed the top-performing group, significantly outperformed

DT, LR, and kNN. Taken together, these findings — supported by confidence intervals and statistical testing — demonstrated that XGBoost provides the best balance between sensitivity and precision, justifying our choice.

These observations indicate that the advantages of XGBoost extend beyond its resilience to missing values. EEG signals are characterized by high temporal variability, moderate signal-to-noise ratios, and strong inter-channel correlations due to volume conduction and neural synchrony. Such properties often degrade the performance of linear models like logistic regression, which cannot capture nonlinear dependencies, and distance-based classifiers such as kNN, which are highly sensitive to redundant or noisy features. XGBoost, as an ensemble of boosted decision trees, is better aligned with these challenges: it models nonlinear feature interactions, selectively splits on informative electrodes, and uses regularization ( $L_1$ ,  $L_2$ , shrinkage, and subsampling) to suppress spurious variance from noisy channels. The ability to exploit correlations between electrodes without being dominated by them likely explains its relative gains over baselines.

Importantly, these same dataset characteristics help explain the model's observed stability across multiple cross-validation strategies.

**Table 7**

Comparison of the proposed model with existing studies on the UCI EEG Eye State Dataset.

| Study                 | Model(s)                           | Best accuracy (%) | Key Highlights                                                                                                                   | Explainability & Interpretability | Missing Data Tolerance |
|-----------------------|------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------------------|
| [12]                  | kNN (k=3), MLP (7 neurons)         | kNN: 84.06        | kNN outperformed MLP; no preprocessing applied                                                                                   | ✗                                 | ✗                      |
| [13]                  | LVQ + BT, CART, LDA, MLP           | LVQ+BT: 94.31     | Hybrid model significantly outperformed baselines in both accuracy and speed                                                     | ✗                                 | ✗                      |
| [10]                  | KStar, 42 classifiers tested       | KStar: 97.30      | Achieved high accuracy; however, processing time was long (approx. 38 min)                                                       | ✗                                 | ✗                      |
| [14]                  | MLP, DBNs, Dropout Networks        | Up to 99.93       | Highest reported accuracy with real-time deep learning model for robust BCI applications                                         | ✗                                 | ✗                      |
| <b>Proposed Model</b> | <b>Tuned XGBoost + SHAP + LIME</b> | <b>95.73</b>      | High accuracy with minimal preprocessing, native handling of missing values, and integrated interpretability using SHAP and LIME | ✓                                 | ✓                      |

Because EEG electrodes provide overlapping and partially redundant information, XGBoost can still form robust decision boundaries even when folds differ in their specific composition of instances. Its boosting mechanism further emphasizes consistent patterns while down-weighting noisy or unstable samples, which reduces variance in performance across data partitions. This interaction between the statistical structure of EEG data (correlations, noise levels) and XGBoost's ensemble learning strategy provides a plausible explanation for the consistently high accuracy and AUC observed under 3-, 5-, 7-, and 10-fold validation as well as the 80/20 split.

Hyperparameter optimization was conducted using grid search in Fig. 5, exploring combinations of max\_depth, n\_estimators, and subsample. The best-performing configuration — max\_depth = 8, n\_estimators = 500, and subsample = 0.8 — achieved a cross-validated accuracy of 95.73%. This configuration was then fixed for all subsequent evaluations.

To validate the model's robustness, we applied multiple cross-validation strategies. Across 3-fold, 5-fold, 7-fold, 10-fold, and an 80/20 train-test split, the model consistently demonstrated strong and stable performance. Accuracy values ranged from 94.68% to 96.23%, with comparably high values for precision, recall, F1 score, and AUC. This confirmed that our model generalized well across different data partitions, reinforcing its reliability.

Next, we simulated real-world challenges by introducing missing values randomly across the dataset in increasing proportions—1%, 3%, 5%, 10%, and 20%. Despite the absence of data imputation, XGBoost handled these scenarios remarkably well. Even with 20% of values missing, the model achieved 87.42% accuracy and an AUC of 94.65%. This resilience stems from XGBoost's built-in ability to learn optimal splits in the presence of missing data, making it exceptionally suitable for deployment in real-world applications where data quality can be inconsistent.

Table 7 compares the proposed XGBoost-based model with several existing approaches applied to the UCI EEG Eye State dataset. While some studies, such as [14], report exceptionally high accuracies (up to 99.93%), they lack critical components such as explainability and robustness to missing data. Notably, none of the existing models offer deep interpretability or proven resilience to incomplete EEG signals, which are both essential for real-world biomedical applications. In contrast, our proposed model not only achieves a competitive accuracy of 95.73% — outperforming many traditional classifiers — but also integrates both global and local explainability methods (SHAP and LIME) and natively tolerates missing data without requiring imputation. These qualities make our approach more transparent, clinically trustworthy, and operationally resilient, demonstrating that interpretability and reliability can coexist with strong predictive performance, even if not the absolute highest accuracy is attained.

### 5.1. Sources and characteristics of misclassification

A comprehensive error analysis was conducted to examine the patterns of misclassified EEG samples and determine whether these errors stemmed from ambiguous signals, overlapping feature spaces, or model limitations. The confusion matrix in Fig. 6 revealed that the classifier correctly identified the majority of samples, with only a small proportion of *Open* states being predicted as *Closed* and vice versa. These errors suggest that a subset of signals exhibited transitional or mixed characteristics, making them less distinguishable for the model.

To better understand the distribution of classifier confidence, the predicted probabilities were analyzed in Fig. 7. As shown, most samples were classified with very high certainty, clustering near probabilities of 0.0 and 1.0. This trend is quantified in Table 8, which reports the distribution of correct and misclassified samples across probability bins. Over 7200 samples fell into the lowest bin (0.00–0.05) and more than 5600 into the highest bin (0.95–1.00), both with very few misclassifications. By contrast, the central bins (0.20–0.55) contained a higher proportion of errors, with misclassification rates exceeding 30%–40% in some intervals. This indicates that ambiguous or borderline EEG signals — those lying close to the decision threshold — are the main contributors to classification errors.

The reliability of predicted probabilities was further validated using the calibration curve in Fig. 8. The near-linear alignment of the curve with the reference diagonal demonstrates that the classifier's probability estimates correspond closely to the observed frequencies. Thus, the observed errors are not the result of miscalibrated outputs but rather reflect genuine signal overlap and uncertainty in the data itself.

Hardness analysis offered additional insights into which samples were consistently difficult to classify across repeated stratified folds. Fig. 9 and Table 9 show that the vast majority of samples (13,547) were never misclassified, confirming the model's robustness. A smaller subset (371 samples) was misclassified only once, while progressively fewer samples accumulated multiple errors. Notably, only 227 samples were misclassified in all 10 repetitions, identifying them as “hard” cases likely arising from noisy or transitional EEG activity.

Taken together, the evidence from Figs. 6–9 and Tables 8–9 suggests that most misclassifications stem from intrinsic ambiguity in EEG signals and overlaps between eye-open and eye-closed patterns. The model itself was both well-calibrated and highly confident in the majority of predictions, implying that systematic limitations of the classifier were not the primary source of error. Instead, errors were concentrated in samples near the decision boundary, which naturally reflect blended neural activity. Future research could address this by incorporating temporal context or feature-level regularization to mitigate uncertainty in borderline cases.

**Table 8**  
Probability distribution of correct vs. Misclassified samples across bins.

| Bin range | Correct samples | Misclassified samples | Total samples |
|-----------|-----------------|-----------------------|---------------|
| 0.00–0.05 | 7183            | 84                    | 7267          |
| 0.05–0.10 | 265             | 39                    | 304           |
| 0.10–0.15 | 145             | 31                    | 176           |
| 0.15–0.20 | 98              | 35                    | 133           |
| 0.20–0.25 | 60              | 21                    | 81            |
| 0.25–0.30 | 63              | 32                    | 95            |
| 0.30–0.35 | 49              | 28                    | 77            |
| 0.35–0.40 | 42              | 20                    | 62            |
| 0.40–0.45 | 37              | 30                    | 67            |
| 0.45–0.50 | 28              | 32                    | 60            |
| 0.50–0.55 | 34              | 25                    | 59            |
| 0.55–0.60 | 39              | 28                    | 67            |
| 0.60–0.65 | 45              | 22                    | 67            |
| 0.65–0.70 | 42              | 21                    | 63            |
| 0.70–0.75 | 74              | 23                    | 97            |
| 0.75–0.80 | 64              | 24                    | 88            |
| 0.80–0.85 | 101             | 26                    | 127           |
| 0.85–0.90 | 146             | 23                    | 169           |
| 0.90–0.95 | 257             | 27                    | 284           |
| 0.95–1.00 | 5569            | 68                    | 5637          |

**Table 9**  
Distribution of samples by misclassification count (Repeated 10-Fold  $\times 10$  Repeats).

| Misclassification count | Number of samples |
|-------------------------|-------------------|
| 0                       | 13547             |
| 1                       | 371               |
| 2                       | 190               |
| 3                       | 141               |
| 4                       | 92                |
| 5                       | 80                |
| 6                       | 80                |
| 7                       | 81                |
| 8                       | 87                |
| 9                       | 84                |
| 10                      | 227               |

### 5.2. Local explainability

To better understand how individual predictions were made by the model, we employed two complementary local explanation techniques — **LIME** and **SHAP** — on four randomly selected EEG instances: two from the “eyes open” class (Class 0) and two from the “eyes closed” class (Class 1). While both techniques aim to interpret model decisions at the instance level, they approach the task differently: LIME builds a locally interpretable surrogate model to approximate the decision boundary, whereas SHAP uses Shapley values to attribute additive contributions to each feature based on the exact model output.

Figs. 11 and 12 show that both LIME and SHAP consistently highlight key features contributing to “eyes open” predictions. According to LIME (see Fig. 11 and Table 10), thresholds such as  $O1 > 4083.59$ ,  $F8 > 4617.44$ , and  $P7 \leq 4611.79$  emerge as influential indicators. Features like  $P8$ ,  $AF3$ , and  $FC5$  provide additional support, while others such as  $T8$  and  $F3$  have minor contributions.

SHAP plots (Fig. 12 and Table 11) reinforce these findings with large negative contributions for  $P7$  ( $-7.28$  and  $-7.34$ ) and  $O1$  ( $-2.30$  and  $-2.16$ ), strongly supporting Class 0 predictions. Other features such as  $AF4$ ,  $AF3$ , and  $FC5$  also play significant roles, validating their importance as seen in the LIME explanation. The consistency across both techniques enhances confidence in the model’s rationale for classifying “eyes open” instances.

For the “eyes closed” cases, LIME (Fig. 11) identifies  $T8 > 4239.49$ ,  $O1$ ,  $F4$ , and  $T7 \leq 4331.79$  as major contributors, with additional influence from  $AF4$ ,  $AF3$ , and  $FC5$ . These threshold-based conditions support the model’s classification toward Class 1.

The SHAP plot (Fig. 12) provides a more nuanced decomposition. Features such as  $AF3$  ( $+3.29$ ),  $T8$  ( $+1.72$ ), and  $AF4$  ( $+1.63$ ) show

strong positive SHAP values, confirming their role in pushing predictions toward Class 1. Meanwhile, features like  $O1$ ,  $F4$ , and  $FC6$  had negative SHAP values, which act as counterforces in the decision process.

Despite their methodological differences, both LIME and SHAP consistently identify overlapping dominant features.  $P7$ ,  $O1$ ,  $F8$ , and  $AF3$  are frequently observed as influential across both classes and techniques. The convergence in feature importance and directionality supports the interpretability of the model and reveals that EEG signals from the temporal and frontal regions are highly informative for eye state classification.

Overall, LIME offers an intuitive, rule-based understanding through threshold-based feature conditions, while SHAP provides a mathematically grounded decomposition of model output. Used together, these techniques deliver complementary insights into how the model interprets EEG data to distinguish between eye states. A direct comparison of their outputs reveals both convergence and divergence: across both methods, electrodes  $P7$ ,  $O1$ ,  $F8$ , and  $AF3$  consistently emerge as dominant contributors, underscoring the robustness of their role in eye state discrimination. At the same time, LIME tends to emphasize frontal and temporal features (e.g.,  $F7$ ,  $T8$ ) in threshold-based rules for “eyes closed” states, whereas SHAP more strongly quantifies the positive contributions of  $AF3$  and  $AF4$  in those same cases. This pattern suggests that LIME is more sensitive to localized, conditional thresholds, while SHAP captures broader additive contributions. The convergence on key posterior features and divergence in frontal-temporal emphasis together enhance confidence in the model’s interpretability while revealing subtle context-dependent aspects of EEG-based eye state classification.

### 5.3. Global explainability

We analyzed the feature importance scores from the tuned XGBoost classifier. As shown in both the bar plot Fig. 10 and the ranked Table 12, the  $P7$  electrode held the highest importance score at 0.1204, followed by  $O1$  (0.0938) and  $F7$  (0.0901). These electrodes corresponded to the parietal, occipital, and frontal regions of the brain, respectively—regions strongly associated with visual processing and attention regulation. Their prominence suggests that changes in eye state (open vs. closed) are largely reflected in these neural areas.

Frontal electrodes such as  $F8$ ,  $AF3$ ,  $F4$ , and  $AF4$  also showed substantial contributions, reinforcing the involvement of the frontal cortex in alertness monitoring and executive attention. Features from the central and temporal regions — including  $FC6$ ,  $T8$ ,  $T7$ , and  $FC5$

**Table 10**  
LIME feature values for two instances per class.

| Instance                 | Top LIME Features with Thresholds                                                                                                                                                                                                                    |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eyes Open – Instance 1   | $O1 > 4083.59$ , $F8 > 4617.44$ , $F7 > 4347.18$ , $P7 \leq 4611.79$ ,<br>$P8 > 4209.23$ , $4276.92 < F4 \leq 4287.18$ , $O2 \leq 4624.10$ ,<br>$AF3 > 4311.79$ , $F8 \leq 4209.66$ , $T3 > 4270.77$ , $FC5 > 4132.31$ ,<br>$AF4 > 4372.82$          |
| Eyes Open – Instance 2   | $O1 > 4083.59$ , $F8 > 4617.44$ , $P8 > 4209.23$ ,<br>$3990.77 < F2 \leq 4005.64$ , $O2 > 4624.10$ , $T8 > 4239.49$ ,<br>$FC5 > 4132.31$ , $AF4 > 4372.82$ , $4200.51 < FC6 \leq 4211.28$ ,<br>$4338.97 < T7 \leq 4347.18$ , $F3 > 4270.77$          |
| Eyes Closed – Instance 1 | $T8 > 4239.49$ , $4070.26 < O1 \leq 4083.59$ , $F4 > 4287.18$ ,<br>$F8 > 4617.44$ , $T7 \leq 4331.79$ , $O2 > 4624.10$ , $P7 \leq 4611.79$ ,<br>$FC6 > 4211.28$ , $P8 > 4209.23$ , $AF4 > 4372.82$ , $F3 > 4270.77$ ,<br>$AF3 > 4311.79$             |
| Eyes Closed – Instance 2 | $T8 > 4239.49$ , $4070.26 < O1 \leq 4083.59$ , $F8 > 4617.44$ ,<br>$T7 \leq 4331.79$ , $3990.77 < F2 \leq 4005.64$ , $O2 > 4624.10$ ,<br>$P7 \leq 4611.79$ , $AF4 > 4372.82$ , $4108.21 < FC5 \leq 4120.51$ ,<br>$FC5 \leq 4132.31$ , $F3 > 4270.77$ |

**Table 11**  
Top SHAP feature contributions for two instances per class.

| Class                 | Instance   | Top 5 features with SHAP values                                        |
|-----------------------|------------|------------------------------------------------------------------------|
| Eyes Open (Class 0)   | Instance 1 | $P7: -7.28$ , $O1: -2.30$ , $AF4: -1.50$ , $AF3: -1.12$ , $FC3: -1.02$ |
|                       | Instance 2 | $P7: -7.34$ , $FC5: -2.16$ , $AF4: -1.84$ , $AF3: -1.31$ , $T8: -0.65$ |
| Eyes Closed (Class 1) | Instance 1 | $AF3: +1.81$ , $AF4: +1.63$ , $F7: +1.47$ , $FC5: -1.30$ , $F8: -1.24$ |
|                       | Instance 2 | $AF3: +3.29$ , $T8: +1.72$ , $F8: -1.44$ , $FC5: -1.25$ , $F7: +1.14$  |

**Table 12**  
Top EEG features ranked by importance.

| Feature | Importance |
|---------|------------|
| P7      | 0.1204     |
| O1      | 0.0938     |
| F7      | 0.0901     |
| F8      | 0.0830     |
| AF3     | 0.0722     |
| F4      | 0.0713     |
| AF4     | 0.0711     |
| F3      | 0.0677     |
| O2      | 0.0647     |
| FC6     | 0.0607     |
| T8      | 0.0602     |
| T7      | 0.0591     |
| FC5     | 0.0526     |
| P8      | 0.0333     |

**Table 13**  
Top EEG Features by Mean Absolute SHAP Values.

| Feature | Mean(  SHAP value  ) |
|---------|----------------------|
| O1      | 1.95                 |
| P7      | 1.80                 |
| F7      | 1.65                 |
| AF3     | 1.25                 |
| F8      | 1.05                 |
| F4      | 0.95                 |
| O2      | 0.90                 |
| AF4     | 0.85                 |
| FC6     | 0.80                 |
| T8      | 0.75                 |
| T7      | 0.70                 |
| FC5     | 0.65                 |
| F3      | 0.60                 |
| P8      | 0.55                 |

— had moderate importance, suggesting they offer complementary information useful for eye state classification.

The least contributing feature was P8, with an importance score of 0.0333, indicating minimal influence from the right parietal area in this dataset. Overall, the spatial distribution of top-ranking features aligned well with the neurological understanding of visual attention and ocular activity regulation.

To further interpret the behavior of the trained XGBoost model, we applied aggregated SHAP analysis, a model-agnostic approach that quantifies the contribution of each feature to the prediction output. According to the computed mean absolute SHAP values (Table 13 and Fig. 13), the most influential features were O1, P7, and F7, with average SHAP values of approximately 1.95, 1.80, and 1.65, respectively. These electrodes are positioned over the occipital, parietal, and frontal regions of the brain—areas well known for their involvement in visual processing, attention control, and eye movement regulation.

Notably, these SHAP findings align closely with the results from the earlier XGBoost feature importance analysis, which also ranked P7, O1, and F7 among the top contributors. While XGBoost importance is derived from how frequently a feature is used to split decision trees, SHAP measures the average magnitude of influence on individual predictions.

The convergence between these two distinct interpretability methods reinforces the reliability of the identified key features and validates their role in EEG-based eye state classification.

In addition to the top three features, SHAP also highlights the importance of AF3, F8, and F4, which are located in the frontal lobe and are associated with cognitive alertness and drowsiness monitoring. Features such as O2, AF4, FC6, and T8 exhibit moderate influence, suggesting that temporal and central-frontal regions contribute secondarily to classification performance. The least contributing feature, P8, shows a relatively small SHAP value of approximately 0.55. By clearly distinguishing between local (illustrative, instance-level) and global (dataset-level) explanations, we ensure that the reported electrode importance patterns are robust and reflect generalizable trends across the full dataset, not just a few selected cases.

#### 5.4. Robustness to missing data

To evaluate the robustness of our trained XGBoost model under real-world conditions, we conducted a targeted missing data experiment. Specifically, we introduced missingness into the five most important EEG features — O1, P7, F7, AF3, and F8 — identified via SHAP and



**Table 14**

Performance under targeted deletion in the top 5 most important EEG features.

| Metric        | 1%    | 3%    | 5%    | 10%   | 20%   |
|---------------|-------|-------|-------|-------|-------|
| Accuracy (%)  | 95.19 | 94.79 | 94.29 | 93.34 | 91.54 |
| Precision (%) | 95.15 | 94.76 | 94.19 | 93.14 | 91.22 |
| Recall (%)    | 94.10 | 93.57 | 93.01 | 91.92 | 89.80 |
| F1 Score (%)  | 94.62 | 94.16 | 93.59 | 92.53 | 90.49 |
| AUC           | 99.04 | 98.89 | 98.70 | 98.29 | 97.43 |

**Table 15**

Performance under random missing values across all EEG features.

| Metric        | 1%    | 3%    | 5%    | 10%   | 20%   |
|---------------|-------|-------|-------|-------|-------|
| Accuracy (%)  | 94.84 | 93.83 | 93.02 | 90.42 | 87.42 |
| Precision (%) | 94.79 | 93.72 | 92.96 | 90.47 | 87.38 |
| Recall (%)    | 93.65 | 92.44 | 91.37 | 87.92 | 84.11 |
| F1 Score (%)  | 94.22 | 93.07 | 92.15 | 89.18 | 85.71 |
| AUC           | 98.90 | 98.46 | 98.13 | 96.86 | 94.65 |

feature importance analysis. Increasing proportions of values (1%, 3%, 5%, 10%, and 20%) were randomly replaced with Not a Number (NaN), simulating typical failures in EEG acquisition such as electrode detachment or sensor noise.

*Performance under targeted deletion.* As shown in Table 14 and Fig. 15, the model exhibits remarkable tolerance to the loss of critical inputs. With only 1% missing values in the top five channels, the model maintains an accuracy of 95.19% and an AUC of 99.04%. Even under extreme conditions — 20% missing data in those key features — accuracy and AUC drop only modestly to 91.54% and 97.43%, respectively. Precision, recall, and F1 score follow similar trends, staying above 90% for all metrics until the highest level of deletion.

*Comparison with random missingness.* To contextualize these findings, we also evaluate the model under randomly distributed missing values across all EEG features (Table 15 and Fig. 14). While performance declined more sharply in this scenario — accuracy drops to 87.42% and AUC falls to 94.65% at 20% missingness — the model still maintains respectable diagnostic metrics, demonstrating robustness even without imputation.

*Insights and practical implications.* These findings highlight two key strengths of the proposed approach. First, the XGBoost classifier inherently manages missing values by routing them through learned default directions in its decision trees, eliminating the need for imputation. Second, the model leverages inter-channel redundancy—when one high-importance channel becomes unavailable, it compensates using informative patterns from correlated electrodes. This fault-tolerant behavior is especially valuable in real-world EEG monitoring, where electrode detachment, motion artifacts, or transient noise are unavoidable, particularly in mobile or long-duration settings. Even under substantial data loss from critical brain regions, the model continues to deliver stable and reliable performance, reducing dependence on heavy preprocessing or repair pipelines.

The quantitative trends provide further insight into this robustness. Under random missingness, accuracy declines by approximately 0.39% for every 1% of additional data loss, whereas targeted deletion of the five most important electrodes (O1, P7, F7, AF3, F8) produces a gentler decline of about 0.19% per 1%. This indicates that diffuse random noise erodes global signal quality more severely than localized loss in critical regions, where redundancy helps the model adapt. Moreover, interaction effects are evident: simultaneous degradation of occipital and parietal electrodes (O1 and P7) lead to sharper declines than when frontal electrodes were affected alone, underscoring the disproportionate contribution of posterior visual-attentional regions to eye state decoding.

Taken together, these results highlight both the practical robustness of the model and its neurophysiological plausibility. The system not only withstands realistic signal disruptions but also reflects known

brain–eye dynamics, where occipital inputs drive visual processing and parietal areas coordinate attentional shifts, forming the backbone of reliable eye state classification.

### 5.5. Neurophysiological interpretation of explainability findings

Integrating the findings from all applied explainability techniques — including XGBoost feature importance, global SHAP values, local SHAP explanations, and LIME — provides a coherent and biologically plausible view of how the model interprets EEG signals for eye state classification. The convergence on specific EEG electrodes across methods enhances confidence in the interpretability and neuroscientific grounding of the model’s decision process.

*Dominant features in visual and attentional processing.* Electrodes O1, P7, and F7 were consistently ranked among the most influential features across all analyses. O1, located over the left occipital lobe, overlays the primary visual cortex and directly responds to retinal input. When the eyes close, visual input to the occipital cortex is suppressed, leading to characteristic increases in alpha activity, which explains why O1 shows strong discriminative power between states. Similarly, P7, positioned over the left parietal cortex, contributes to visuospatial attention and sensory integration. Its importance likely reflects the parietal cortex’s role in reallocating attentional resources when external visual input is absent. The prominence of F7, situated over the left frontal region, suggests involvement in prefrontal modulation of visual attention and alertness, consistent with its role in monitoring behavioral states such as vigilance and drowsiness.

*Frontal lobe contributions to drowsiness and alertness.* Frontal electrodes including AF3, F8, AF4, and F4 also exhibited strong importance in both global and local explainability frameworks. These regions overlay the prefrontal cortex, known for its role in cognitive control, wakefulness monitoring, and executive functioning. Notably, in local SHAP and LIME explanations for “eyes closed” instances, features like AF3 and AF4 demonstrated high positive contributions. This aligns with evidence that eye closure shifts neural activity toward internally directed attention and reduced sensory processing, reflecting increased frontal theta rhythms linked to drowsiness.

*Moderate but meaningful contributions from temporal and central regions.* Electrodes such as T7, T8, FC5, and FC6 consistently appeared with moderate importance across interpretability methods. These sites correspond to the temporal and central-frontal regions, which are involved in sensory integration, auditory processing, and secondary visual functions. In particular, T8 showed elevated SHAP contributions in “eyes closed” instances, possibly reflecting recruitment of memory- and imagery-related processes that are more active when visual input is suppressed.

**Low contribution of the right parietal region.** In contrast, the electrode P8, located over the right parietal lobe, was consistently ranked among the least important features across all global and local analyses. Its low impact, compared to the strong influence of its left-hemisphere counterpart P7, may suggest a lateralization effect. This asymmetry aligns with findings that left parietal regions are more actively engaged in sustained attention tasks involving eye state, while right parietal contributions emerge more prominently in visuospatial exploration rather than sustained vigilance.

**Summary of neurophysiological relevance.** Overall, the agreement among different interpretability techniques not only supports the reliability of the model but also validates its neurophysiological plausibility. The high importance of occipital (O1), parietal (P7), and frontal (F7, AF3/AF4, F8) electrodes directly maps to the known dynamics of eye closure: reduced visual input to occipital cortex, redistribution of attention via parietal networks, and increased frontal monitoring of internal versus external states. This alignment reinforces the potential for EEG-based eye state classification models to offer interpretable insights in both clinical and cognitive monitoring applications.

## 6. Limitations

While the proposed framework demonstrated strong performance, several limitations should be acknowledged.

- **Single dataset dependency:** The study relied on a single dataset (UCI EEG Eye State), which is limited in terms of subject diversity, short recording duration (117 s), and acquisition conditions. As such, the generalizability of the findings to other populations or recording setups remains uncertain.
- **Potential dataset-specific overfitting:** Although multiple validation strategies (3-fold, 5-fold, 7-fold, 10-fold, and 80/20 split) were employed to reduce the likelihood of overfitting, the high reported metrics may still partially reflect dataset-specific characteristics rather than true population-level performance.
- **EEG artifact contamination:** EEG recordings are inherently prone to contamination from artifacts such as muscle activity, eye blinks, and electrode noise. While XGBoost's resilience to missing or corrupted values provided some tolerance, explicit artifact detection and removal were not addressed in this study.
- **Simplified missing-data assumption:** Missing values were simulated under a MCAR assumption to enable controlled robustness analysis. However, real-world EEG data loss is often structured and correlated due to electrode detachment or motion artifacts. Therefore, the reported robustness results should be interpreted within the scope of MCAR conditions, and future work will explore more realistic missingness mechanisms.
- **Class imbalance effects:** While the dataset exhibits only a mild class imbalance (55.1% open eyes vs. 44.9% closed eyes), this imbalance could still influence the learned decision boundaries under different sampling scenarios.
- **Limited preprocessing:** Advanced pre-processing pipelines (e.g., Independent Component Analysis, wavelet denoising, or artifact-aware rejection) were not implemented. As a result, residual artifacts may still influence classification outcomes.
- **Absence of raw continuous EEG signals:** The dataset provides pre-structured samples rather than continuous EEG waveforms, which prevents direct computation of frequency-domain features such as alpha-band power. This limits the ability to exploit well-established physiological markers of eye-closure and eye-opening conditions, and constrains the analysis to electrode-level values only.
- **Lack of spectral and complexity-based features:** This study operates at the electrode-domain level and does not incorporate

explicit spectral or signal-complexity features, such as frequency-specific power or entropy-based representations that are commonly used in neuroscience-oriented EEG analysis. Consequently, the proposed framework emphasizes spatial and amplitude-based signal characterization rather than frequency-domain or complexity-driven physiological markers.

- **Scope of explainability interpretations:** Both local and global SHAP and LIME analyses were applied; however, given the pre-processed, single-subject nature of the dataset, the selection of representative instances and visualization thresholds remains heuristic, and global electrode importance patterns should be interpreted as exploratory rather than population-level conclusions.
- **Dataset-specific explainability patterns:** While SHAP and LIME offered valuable interpretability, their electrode importance patterns are dataset-specific and should be validated on larger, multi-subject corpora.

These limitations highlight the need for future work to evaluate the proposed approach on larger, more diverse, and artifact-rich datasets, to incorporate dedicated preprocessing pipelines for artifact mitigation, and to confirm the stability of explainability findings across different populations and recording devices.

## 7. Future directions

Building upon the strengths of the current framework, several promising avenues for future research are identified:

- **Extension to multi-class and continuous prediction:** Move beyond binary eye state classification to detect finer states such as blinking, squinting, or drowsiness levels. Alternatively, model eye openness as a continuous variable to enable more granular behavioral analysis.
- **Deployment in real-time environments:** Integrate the model into real-time applications such as driver monitoring systems, smart eyewear, or wearable brain-computer interface (BCI) devices to evaluate latency, stability, and performance under real-world conditions.
- **Cross-dataset generalization via transfer learning:** Explore transfer learning and domain adaptation strategies to improve model generalization across different EEG datasets, subject populations, and acquisition hardware, reducing the need for retraining.
- **Enhanced neurophysiological interpretation:** Utilize explainability outputs to investigate subject-specific patterns and inter-individual variability, contributing to deeper neuroscientific insights into visual attention and cortical dynamics.
- **Validation of explainability across multi-subject EEG datasets:** Future work should validate both local and global explainability findings on larger, multi-subject EEG datasets, with systematically defined instance selection and thresholding strategies, to assess the robustness and generalizability of electrode importance patterns beyond exploratory analysis.
- **Multimodal integration:** Augment the EEG-based model with complementary biosignals — such as electrooculography (EOG), electrocardiography (ECG), facial landmarks, or video-based cues — to enhance robustness and contextual understanding.
- **Validation on large-scale, noisy, and imbalanced datasets:** Evaluate the model's scalability and fault tolerance on more diverse datasets, incorporating naturally occurring noise and class imbalance to simulate real-world deployment scenarios.
- **Artifact-aware preprocessing and analysis:** Future experiments must incorporate dedicated pipelines for detecting and mitigating EEG artifacts such as muscle activity, ocular movements, and electrode noise. Combining preprocessing methods (e.g., Independent Component Analysis, wavelet denoising) with model-level robustness testing would provide a more comprehensive evaluation of performance under realistic recording conditions.

- **Model robustness under structured and correlated missing EEG data:** Future work will extend the current robustness analysis beyond Missing Completely at Random assumptions by investigating structured and correlated missing-data patterns, such as region-specific electrode dropout or sensor detachment. This would enable a more realistic assessment of model resilience under practical EEG acquisition and deployment conditions.
- **Hybrid deep learning with preprocessing:** Future work could explore combining deep learning architectures (e.g., Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs)) with explainability frameworks and artifact-aware preprocessing methods (e.g., Independent Component Analysis (ICA), band-pass filtering, wavelet denoising). Such hybrid pipelines may further enhance generalizability, although they would require explicit strategies for handling missing data, which is a key strength of the present XGBoost-based approach.
- **Exploration of advanced hyperparameter optimization:** While this study employed Grid Search due to its transparency and the manageable size of the parameter space, future research could investigate more advanced optimization methods such as Random Search, Bayesian Optimization, or evolutionary algorithms. These approaches may improve efficiency and scalability when exploring larger or higher-dimensional hyperparameter spaces, and could provide further insights into the stability and robustness of model configurations.
- **Comparison with frequency- and complexity-based EEG classifiers:** Future work will extend the evaluation to classifiers that leverage frequency-domain and signal-complexity features (e.g., band-power, entropy, and complexity measures), enabling systematic comparison with conventional signal-processing-driven approaches in terms of performance, interpretability, and practical suitability for neuroscience applications.
- **Incorporation of frequency-domain analysis through new data collection:** To enable spectral feature extraction, future studies should consider repeating similar eye state experiments under comparable recording settings while storing continuous raw EEG signals. This would make it possible to calculate frequency-domain measures, such as alpha band (8–12 Hz) power increases during eyes-closed conditions, and integrate them with classifier-based analyses.
- **Mitigation of ambiguous and hard-to-classify samples:** Building on the error analysis conducted in this study, future work should focus on strategies to reduce misclassifications stemming from ambiguous EEG signals and overlapping feature distributions. Potential approaches include incorporating uncertainty-aware models (e.g., Bayesian variants of XGBoost or ensemble learning), adopting cost-sensitive learning to address borderline cases, and implementing a reject option for predictions near the decision threshold (e.g., 0.45–0.55). Such methods would enhance reliability in real-world applications by ensuring that uncertain cases are flagged rather than misclassified.

## 8. Conclusion

This study presented a robust, interpretable, electrode-domain framework for EEG eye state classification using XGBoost on voltage time-series. The model achieved 95.73% accuracy and 99.13% AUC, outperforming baselines across validation schemes and demonstrating that reliable discrimination can be achieved without handcrafted spectral features, leveraging the discriminative power of spatial EEG patterns. A core contribution is the framework's resilience to signal degradation. Through systematic evaluation under realistic channel dropout — including random and targeted removal of key electrodes (O1, P7, F7) — we demonstrated graceful degradation, maintaining stable performance even with 20% missing data without imputation. This fault tolerance makes it suitable for practical EEG use where

poor SNR, artifacts, or montage limitations occur. For physiologically grounded interpretation, complementary explainability methods (XGBoost importance, SHAP, LIME) consistently highlighted dominant contributions from frontal, parietal, and occipital electrodes. This aligns with the known neurophysiology of eye closure (visual processing, attentional modulation, alpha-rhythm dynamics), providing transparent, signal-aware insight that builds trust for clinical use. Unlike prior work focusing solely on accuracy, we show that robust performance, channel-loss resilience, and physiological explainability can coexist in an electrode-domain framework. The approach thus offers a streamlined, signal-processing-oriented alternative to spectral or connectivity-based pipelines, providing a reliable foundation for clinical monitoring, BCIs, and other real-world EEG applications where performance under non-ideal conditions is critical.

## CRedit authorship contribution statement

**Hamzah Dabool:** Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Zaher Aghbari:** Supervision, Conceptualization. **Naveed Ahmed:** Writing – review & editing, Conceptualization. **Imad Afyouni:** Writing – review & editing, Conceptualization. **Hany Alashwal:** Supervision. **Shifan Khanday:** Writing – review & editing, Supervision.

## Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used OpenAI ChatGPT in order to proofread and check for grammatical and punctuation errors and enhance some sentences. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix. Mathematical formulations of explainability methods

### A.1. Evaluation metrics

In this subsection, we provide the mathematical definitions of the evaluation metrics used in our study. These include Accuracy, Precision, Recall, F1 Score, and AUC (Area Under the ROC Curve).

#### A.1.1. Accuracy

Accuracy measures the proportion of correct predictions among all predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (\text{A.1})$$

#### A.1.2. Precision

Precision quantifies the proportion of true positive predictions among all predicted positives:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (\text{A.2})$$

#### A.1.3. Recall (Sensitivity)

Recall, also known as Sensitivity or True Positive Rate, measures the ability to identify actual positives:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (\text{A.3})$$

#### A.1.4. F1 Score

F1 Score is the harmonic mean of Precision and Recall, balancing the two metrics:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (\text{A.4})$$

#### A.1.5. AUC (Area under the ROC curve)

AUC evaluates the model's ability to discriminate between classes across all thresholds:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d\text{FPR} \quad (\text{A.5})$$

### A.2. Feature importance (Gain criterion in XGBoost)

XGBoost measures feature importance based on the improvement in the model's objective function when a feature is used to split a node. This improvement, called *Gain*, can be derived from the regularized objective function and the second-order Taylor expansion of the loss. Below are the main steps.

First, for a given tree at iteration  $t$ , XGBoost approximates the loss using a second-order Taylor expansion:

$$\mathcal{L}^{(t)} \simeq \sum_{i=1}^n \left[ g_i w_{q(x_i)} + \frac{1}{2} h_i w_{q(x_i)}^2 \right] + \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (\text{A.6})$$

where  $g_i = \frac{\partial \ell(y_i, \hat{y}_i)}{\partial \hat{y}_i}$  is the gradient,  $h_i = \frac{\partial^2 \ell(y_i, \hat{y}_i)}{\partial \hat{y}_i^2}$  is the Hessian,  $w_j$  is the weight of leaf  $j$ ,  $T$  is the number of leaves, and  $\lambda, \gamma$  are regularization parameters.

For a node with instance set  $I$ , the aggregated gradient and Hessian are:

$$G = \sum_{i \in I} g_i, \quad H = \sum_{i \in I} h_i \quad (\text{A.7})$$

The optimal weight minimizing Eq. (A.6) for a leaf is:

$$w^* = -\frac{G}{H + \lambda} \quad (\text{A.8})$$

Its corresponding score is:

$$\mathcal{L}_{leaf} = -\frac{1}{2} \cdot \frac{G^2}{H + \lambda} \quad (\text{A.9})$$

When splitting into left ( $L$ ) and right ( $R$ ) children, the improvement in loss is:

$$\text{Gain} = \frac{1}{2} \left( \frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right) - \gamma \quad (\text{A.10})$$

Finally, the overall importance of a feature  $f$  is the sum of gains over all splits where  $f$  is used:

$$\text{FeatureImportance}(f) = \sum_{\text{splits on } f} \text{Gain}_{split} \quad (\text{A.11})$$

This process ensures that features which contribute most to reducing the model's loss across all trees are ranked as most important.

### A.3. LIME (Local interpretable model-agnostic explanations)

For each instance  $x$ , LIME generates a set of perturbed samples  $z' \in Z$  by slightly modifying the original feature values. Each perturbed sample is evaluated using the complex model  $f$ , and the outputs are used as labels for training the surrogate model  $g$ . The proximity between  $z'$  and  $x$  is measured by a kernel function  $\pi_x(z')$ , which assigns higher weights to samples closer to  $x$ .

The local explanation is obtained by solving the following optimization problem:

$$\hat{g} = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \quad (\text{A.12})$$

where:

$$\mathcal{L}(f, g, \pi_x) = \sum_{z' \in Z} \pi_x(z') (f(z') - g(z'))^2 \quad (\text{A.13})$$

is the weighted loss between the predictions of the complex model  $f$  and the surrogate model  $g$ , and

$$\Omega(g) = \text{Complexity penalty of } g \quad (\text{A.14})$$

ensures that  $g$  remains simple (e.g., linear regression with few nonzero coefficients).

Finally, the explanation for instance  $x$  is represented by the coefficients of  $\hat{g}$ , which indicate the contribution of each feature to the local prediction:

$$\text{Explanation}(x) = \{w_1, w_2, \dots, w_M\}, \quad w_i \in \mathbb{R} \quad (\text{A.15})$$

Here,  $w_i$  quantifies how strongly feature  $i$  influences the prediction of  $f$  for instance  $x$ , with positive weights driving the prediction toward one class and negative weights toward the other.

### A.4. SHAP (Shapley additive explanations)

At the local level, SHAP explains an individual prediction  $f(x)$  for a given instance  $x$  by decomposing it into a baseline value  $\phi_0$  and feature-specific contributions  $\phi_i$ :

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i \quad (\text{A.16})$$

where:

- $f(x)$  is the model's prediction for instance  $x$ .
- $\phi_0$  is the base value, typically the expected output of the model over all training data.
- $\phi_i$  is the SHAP value for feature  $i$ , representing its contribution to the prediction.
- $M$  is the total number of features.

The SHAP value for feature  $i$  is computed as the weighted average of its marginal contributions across all possible subsets  $S \subseteq F \setminus \{i\}$  of the feature set  $F$ :

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(M - |S| - 1)!}{M!} [f(S \cup \{i\}) - f(S)] \quad (\text{A.17})$$

Here,  $f(S)$  represents the model prediction using only the subset of features  $S$ . The factorial terms weight each subset fairly, ensuring that  $\phi_i$  reflects the average marginal contribution of feature  $i$  across all possible coalitions of features.

While Eqs. (A.16) and (A.17) provide explanations at the individual instance level (local interpretability), SHAP can be generalized to the global level by aggregating the absolute SHAP values across all instances in the dataset. The global importance of feature  $i$  is given by:

$$\text{Global Importance}(i) = \frac{1}{N} \sum_{j=1}^N |\phi_i^{(j)}| \quad (\text{A.18})$$

where:

- $N$  is the total number of instances in the dataset.
- $\phi_i^{(j)}$  is the SHAP value of feature  $i$  for instance  $j$ .

This aggregation yields a ranked measure of global importance, highlighting which features consistently contribute the most to the model's predictions. Thus, SHAP provides a unified framework that links local interpretability at the instance level to global interpretability across the entire dataset.

## Data availability

Data will be made available on request.

## References

- [1] M. Del Pozo-Banos, J.B. Alonso, J.R. Ticay-Rivas, C.M. Travieso, Electroencephalogram subject identification: A review, *Expert Syst. Appl.* 41 (15) (2014) 6537–6554.
- [2] Y. Zhu, Q. Wang, L. Zhang, Study of EEG characteristics while solving scientific problems with different mental effort, *Sci. Rep.* 11 (1) (2021) 23783.
- [3] P.A.d.M. Kanda, R. Anghinah, M.T. Smidh, J.M. Silva, The clinical use of quantitative EEG in cognitive disorders, *Dement. Neuropsychol.* 3 (3) (2009) 195–203.
- [4] C. Im, J.-M. Seo, A review of electrodes for the electrical brain signal recording, *Biomed. Eng. Lett.* 6 (3) (2016) 104–112.
- [5] R.J. Barry, A.R. Clarke, S.J. Johnstone, C.A. Magee, J.A. Rushby, EEG differences between eyes-closed and eyes-open resting conditions, *Clin. Neurophysiol.* 118 (12) (2007) 2765–2773.
- [6] H. Yaacob, F. Hossain, S. Shari, S.K. Khare, C.P. Ooi, U.R. Acharya, Application of artificial intelligence techniques for brain–computer interface in mental fatigue detection: A systematic review (2011–2022), *IEEE Access* 11 (2023) 74736–74758.
- [7] N.S. Amer, S.B. Belhaouari, EEG signal processing for medical diagnosis, healthcare, and monitoring: A comprehensive review, *IEEE Access* 11 (2023) 143116–143142.
- [8] Y. Shigeta, et al., Diagnosis and treatment for obstructive sleep apnea: Fundamental and clinical knowledge in obstructive sleep apnea, *J. Prosthodont. Res.* 59 (3) (2015) 161–171.
- [9] D. Kan, P. Croarkin, C. Phang, P. Lee, EEG differences between eyes-closed and eyes-open conditions at the resting stage for euthymic participants, *Neurophysiology* 49 (6) (2017) 432–440.
- [10] O. Rösler, D. Suendermann, A first step towards eye state prediction using eeg, *Proc. AIHLS* 1 (2013) 1–4.
- [11] S. Aydın, Cross-validated adaboost classification of emotion regulation strategies identified by spectral coherence in resting-state, *Neuroinformatics* 20 (3) (2022) 627–639.
- [12] M. Koklu, K. Sabancı, The classification of eye state by using kNN and MLP classification models according to the EEG signals, *Int. J. Intell. Syst. Appl. Eng.* 3 (4) (2015) 127–130.
- [13] M. Nilashi, R.A. Abumalloh, H. Ahmadi, S. Samad, A. Alghamdi, M. Alrizq, S. Alyami, F.K. Nayer, Electroencephalography (EEG) eye state classification using learning vector quantization and bagged trees, *Heliyon* 9 (4) (2023).
- [14] T.K. Reddy, L. Behera, Online eye state recognition from EEG data using deep architectures, in: 2016 IEEE International Conference on Systems, Man, and Cybernetics, SMC, IEEE, 2016, pp. 000712–000717.
- [15] M. Naderi, A. Jahanian-Najafabadi, A systematic review of EEG-based machine learning classifications for obsessive-compulsive disorder: current status and future directions, *BMC Psychiatry* 25 (1) (2025) 854.
- [16] O. Roesler, EEG Eye State, 2013, <http://dx.doi.org/10.24432/C57G7J>, UCI Machine Learning Repository.
- [17] T. Chen, T. He, M. Benesty, V. Khotilovich, Y. Tang, H. Cho, K. Chen, R. Mitchell, I. Cano, T. Zhou, et al., Xgboost: extreme gradient boosting, 1, (4) 2015, pp. 1–4, R package version 0.4-2.
- [18] H. Dabool, H. Alashwal, A. Moustafa, Enhancing diagnostic accuracy by bypassing traditional imputation and leveraging missing data in alzheimer's disease detection models, in: Proceedings of the 2024 8th International Conference on Information System and Data Mining, 2024, pp. 33–38.
- [19] H. Dabool, H. Alashwal, H. Alnuaimi, A. Alhouqani, S. Alkaabi, A. Al Ahabbi, Comparative analysis of hyperparameter tuning methods in classification models for ensemble learning, in: 2024 7th International Conference on Algorithms, Computing and Artificial Intelligence, ACAI, IEEE, 2024, pp. 1–5.
- [20] P. Biecek, T. Burzykowski, Local interpretable model-agnostic explanations (LIME), *Explan. Model. Anal. Explor. Explain. Examine Predict. Model.* 1 (2021) 107–124.
- [21] Y. Nohara, K. Matsumoto, H. Soejima, N. Nakashima, Explanation of machine learning models using shapley additive explanation and application for real data in hospital, *Comput. Methods Programs Biomed.* 214 (2022) 106584.